

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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1. INTRODUCTION

Definition 1.1. Let $1 \leq r < \infty$. For a measurable set $E \subset \mathbb{R}^N$ of finite positive measure and $f \in L^r(E)$, define

$$\text{osc}_r(f; E) := \left(\frac{1}{|E|} \int_E |f - (f)_E|^r dx \right)^{1/r}.$$

Definition 1.2. Let $0 < p < \infty$, $0 < q \leq \infty$, and $1 \leq r < \infty$. Define

$$OSC^{p,q}(\mathbb{R}^N) := \left\{ f \in L^r_{\text{loc}}(\mathbb{R}^N) : \|f\|_{OSC^{p,q}(\mathbb{R}^N)} < \infty \right\},$$

where

$$\|f\|_{OSC^{p,q}(\mathbb{R}^N)} := \sum_{v=1}^{2^N} \left\| \{\text{osc}_r(f; Q)\}_{Q \in \mathbb{D}^v} \right\|_{\ell^{p,q}}.$$

By [19, Proposition 4.1], the above definition is independent of r .

Date: September 6, 2026.

2010 Mathematics Subject Classification. 47B10, 42B20, 43A85.

Key words and phrases. Schatten class, Riesz transform commutators, Dunkl operator, Besov space, Nearly weakly orthogonal sequence.

1.1. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N , p , and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda \ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p , which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a measurable set E by

$$(f)_{E,dx} := \frac{1}{|E|} \int_E f(x) dx.$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper. We will denote by $O(x) = \{\sigma(x) : \sigma \in G\}$ the G -orbit of a point $x \in \mathbb{R}^N$, and $O(E) = \bigcup_{x \in E} O(x)$ the G -orbit of $E \subset \mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(2.1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2.2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

[Add the definition of A_∞ . Zhijie will do it.]

Lemma 2.1. *Let*

$$dw(x) = \varpi(x) dx, \quad \varpi(x) = \prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)}.$$

Then $\varpi \in A_\infty(\mathbb{R}^N, dx)$.

Proof. The assertion is immediate if $k \equiv 0$. Otherwise, choose $p > 1$ sufficiently large so that

$$\frac{2|R_+|k(v)}{p-1} < 1, \quad v \in R_+.$$

For every $0 < \beta < 1$, $v \in R_+$, and Euclidean ball $B(x_0, r)$, we apply an orthogonal change of variables to deduce that

$$\begin{aligned} \int_{B(x_0, r)} |\langle x, v \rangle|^{-\beta} dx &\lesssim \frac{1}{r|v|} \int_{\langle x_0, v \rangle - |v|r}^{\langle x_0, v \rangle + |v|r} |s|^{-\beta} ds \\ &= (|v|r)^{-\beta} \int_{\frac{\langle x_0, v \rangle}{|v|r} - 1}^{\frac{\langle x_0, v \rangle}{|v|r} + 1} |s|^{-\beta} ds \\ &\lesssim (|v|r)^{-\beta} \left(1 + \frac{|\langle x_0, v \rangle|}{|v|r} \right)^{-\beta} \\ &\lesssim (|\langle x_0, v \rangle| + r)^{-\beta}. \end{aligned}$$

Hence, by Hölder's inequality,

$$\begin{aligned} \int_{B(x_0, r)} \varpi^{-\frac{1}{p-1}} dx &= \int_{B(x_0, r)} \prod_{v \in R_+} |\langle x, v \rangle|^{-\frac{2k(v)}{p-1}} dx \\ &\leq \prod_{v \in R_+} \left(\int_{B(x_0, r)} |\langle x, v \rangle|^{-\frac{2|R_+|k(v)}{p-1}} dx \right)^{\frac{1}{|R_+|}} \\ &\lesssim \prod_{v \in R_+} (|\langle x_0, v \rangle| + r)^{-\frac{2k(v)}{p-1}}. \end{aligned}$$

On the other hand, by (2.1) and the G -invariance of k ,

$$\int_{B(x_0, r)} \varpi dx = \frac{w(B(x_0, r))}{|B(x_0, r)|} \lesssim \prod_{v \in R_+} (|\langle x_0, v \rangle| + r)^{2k(v)}.$$

Therefore,

$$\left(\int_B \varpi dx \right) \left(\int_B \varpi^{-\frac{1}{p-1}} dx \right)^{p-1} \lesssim 1$$

uniformly over all Euclidean balls B . Thus $\varpi \in A_p(\mathbb{R}^N, dx)$, and hence $\varpi \in A_\infty(\mathbb{R}^N, dx)$. $\square$

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

Definition 2.2. The Weyl chambers are defined as the closures of connected components of the set

$$(2.3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define $d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|$ as the distance between two G -orbits $O(x)$ and $O(y)$. Recall from [10, Chapter VII, proof of Theorem 2.12] that

$$(2.4) \quad d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma$$

for some closed Weyl chamber Γ .

It is clear that $O(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$. Moreover,

$$w(B(x, r)) \leq w(O(B(x, r))) \leq |G|w(B(x, r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$, i.e., $B^d(x, r) = O(B(x, r))$.

Denote by $K_{\text{Dunkl},j}(x, y)$ the integral kernel of $R_{\text{Dunkl},j}$. For every $x, y \in \mathbb{R}^N$ and $r > 0$, let

$$V(x, y, r) := \max\{w(B(x, r)), w(B(y, r))\}.$$

The following lemma establishes derivative estimates for the Dunkl-Riesz transform kernel.

Lemma 2.3. Assume that $N \geq 1$. Then for every pair of multi-indices α, β , every $j \in \{1, \dots, N\}$, and every $x, y \in \mathbb{R}^N$ with $d(x, y) > 0$,

$$(2.5) \quad \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

Proof. **Case 1.** Suppose that $|\alpha| + |\beta| = 0$. By [8, Theorem 1.1],

$$|K_{\text{Dunkl},j}(x, y)| \lesssim \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

By the definition of d , the G -invariance of dw , and (2.2),

$$V(x, y, d(x, y)) \lesssim w(B(x, d(x, y))),$$

and hence (2.5) follows.

Case 2. Suppose that $|\alpha| + |\beta| \geq 1$. By [3, Theorem 3.1],

$$\left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \lesssim_{\alpha, \beta, R, k} t^{-\frac{|\alpha|+|\beta|}{2}} \left(1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t}.$$

Recall from the proof of Lemma 2.4 that

$$K_{\text{Dunkl},j}(x, y) = -\frac{C}{2} \int_0^\infty (y_j - x_j) h_t(x, y) \frac{dt}{t^{3/2}}.$$

Since

$$\partial_x^\alpha \partial_y^\beta ((y_j - x_j) h_t(x, y)) = (y_j - x_j) \partial_x^\alpha \partial_y^\beta h_t(x, y) - \alpha_j \partial_x^{\alpha-e_j} \partial_y^\beta h_t(x, y) + \beta_j \partial_x^\alpha \partial_y^{\beta-e_j} h_t(x, y),$$

where the corresponding term is omitted if $\alpha_j = 0$ or $\beta_j = 0$, we obtain

$$\begin{aligned} |\partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y)| &\lesssim \frac{1}{\|x - y\|} \int_0^\infty t^{-\frac{|\alpha|+|\beta|+1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt \\ &\quad + \frac{1}{\|x - y\|^2} \int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt. \end{aligned}$$

By (2.2),

$$V(x, y, \sqrt{t})^{-1} \lesssim V(x, y, d(x, y))^{-1} \begin{cases} \left(\frac{d(x, y)}{\sqrt{t}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{\sqrt{t}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Hence,

$$\begin{aligned} &\int_0^\infty t^{-\frac{|\alpha|+|\beta|+1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt \\ &\lesssim \frac{1}{V(x, y, d(x, y))} \left(d(x, y)^N \int_0^{d(x,y)^2} t^{-\frac{N+|\alpha|+|\beta|+1}{2}} e^{-cd(x,y)^2/t} dt + d(x, y)^N \int_{d(x,y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|+1}{2}} dt \right) \\ &\lesssim \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))}. \end{aligned}$$

The first integral in the parentheses is estimated by the change of variables $s = d(x, y)^2/t$, while the second one converges since $N + |\alpha| + |\beta| > 1$.

Case 2.1. Suppose that $N + |\alpha| + |\beta| > 2$. We apply the same argument to see that

$$\begin{aligned} &\int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt \\ &\lesssim \frac{1}{V(x, y, d(x, y))} \left(d(x, y)^N \int_0^{d(x,y)^2} t^{-\frac{N+|\alpha|+|\beta|}{2}} e^{-cd(x,y)^2/t} dt + d(x, y)^N \int_{d(x,y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|}{2}} dt \right) \\ &\lesssim \frac{d(x, y)^{2-|\alpha|-|\beta|}}{V(x, y, d(x, y))}. \end{aligned}$$

Since $d(x, y) \leq \|x - y\|$, it follows that

$$|\partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y)| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

Case 2.2. Suppose that $N = 1$ and $|\alpha| + |\beta| = 1$. For the terms in which a derivative falls on $y_j - x_j$, retaining the factor $(1 + \|x - y\|/\sqrt{t})^{-2}$ and splitting the integral at $d(x, y)^2$ and $\|x - y\|^2$, we obtain from (2.2)

$$\int_0^\infty t^{-3/2} \left(1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt$$

$$\begin{aligned}
&\lesssim \frac{d(x, y)^N}{\|x - y\|^2 V(x, y, d(x, y))} \int_0^{d(x, y)^2} t^{-\frac{N+1}{2}} e^{-cd(x, y)^2/t} dt \\
&\quad + \frac{d(x, y)}{\|x - y\|^2 V(x, y, d(x, y))} \int_{d(x, y)^2}^{\|x - y\|^2} \frac{dt}{t} + \frac{d(x, y)}{V(x, y, d(x, y))} \int_{\|x - y\|^2}^{\infty} \frac{dt}{t^2} \\
&\lesssim \frac{1}{\|x - y\| V(x, y, d(x, y))},
\end{aligned}$$

where we used

$$\frac{d(x, y)}{\|x - y\|} \log \frac{\|x - y\|}{d(x, y)} \lesssim 1.$$

Combining this estimate with the preceding estimate for the first term, we obtain

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{1}{\|x - y\| V(x, y, d(x, y))} = \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-1}}{V(x, y, d(x, y))}.$$

This completes the proof of Lemma 2.3. $\square$

Lemma 2.4. *For $j \in \{1, 2, \dots, N\}$, and for every ball $B = B(x_0, r) \subset \mathbb{R}^N$, there is another ball $\widehat{B} = B(y_0, r)$ with $y_0 = x_0 + 5re_j$ such that for every $(x, y) \in B \times \widehat{B}$, $K_{\text{Dunkl}, j}(x, y)$ does not change sign and satisfies*

$$|K_{\text{Dunkl}, j}(x, y)| \geq \frac{C}{w(B(x_0, r))}.$$

Proof. The lower bound is given in [8, Theorem 1.2], while the fact that $K_{\text{Dunkl}, j}(x, y)$ does not change sign can be seen in the proof of [8, Theorem 1.2]. Indeed, with the choice $y_0 := x_0 + 5re_j$, we have $y_j - x_j \geq 3r$ for all $(x, y) \in B \times \widehat{B}$. Together with the fact that the heat kernel $h_t(x, y)$ associated with the Dunkl Laplacian is positive, the representation

$$K_{\text{Dunkl}, j}(x, y) = -\frac{C}{2} \int_0^\infty \frac{y_j - x_j}{t} h_t(x, y) \frac{dt}{\sqrt{t}},$$

implies that $K_{\text{Dunkl}, j}(x, y) < 0$ (and thus, does not change sign) on $B \times \widehat{B}$. $\square$

2.2. Systems of dyadic cubes.

Definition 2.5. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathcal{D} := \bigcup_{k \in \mathbb{Z}} \mathcal{D}_k$, $\mathcal{D}_k := \{Q_\alpha^k : \alpha \in \mathcal{A}_k\}$, of Borel sets $Q_\alpha^k \subseteq X$ is called a *system of dyadic cubes over X* if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha \in \mathcal{A}_k} Q_\alpha^k;$$

(ii) If $\ell \geq k$, then either $Q_\beta^\ell \subseteq Q_\alpha^k$ or $Q_\alpha^k \cap Q_\beta^\ell = \emptyset$;

(iii) For each (k, α) and each $\ell \leq k$, there exists a unique β such that $Q_\alpha^k \subseteq Q_\beta^\ell$;

(iv) For each (k, α) there exist 2^N indices β such that

$$Q_\beta^{k+1} \subseteq Q_\alpha^k, \text{ and } Q_\alpha^k = \bigcup_{\substack{Q \in \mathcal{D}_{k+1} \\ Q \subseteq Q_\alpha^k}} Q;$$

(v) For each (k, α) , one has

$$B_X(x_\alpha^k, A_1 2^{-k}) \subseteq Q_\alpha^k \subseteq B_X(x_\alpha^k, A_2 2^{-k}) =: B(Q_\alpha^k)$$

for some positive constants A_1, A_2 depending only on the geometry of X , where we set $B_X(x, r) := B(x, r) \cap X$ for all $x \in X$ and $r > 0$.

(vi) If $\ell \geq k$ and $Q_\beta^\ell \subseteq Q_\alpha^k$, then

$$B(Q_\beta^\ell) \subseteq B(Q_\alpha^k).$$

Each Q_α^k is called a *dyadic cube of generation k* , with center $x_\alpha^k \in Q_\alpha^k$ and side-length $\ell(Q_\alpha^k) = 2^{-k}$.

Notation 2.6. For every dyadic cube $Q \in \mathcal{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Notation 2.7. Let

$$\mathbb{D}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^0$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1 + 1) 2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N + 1) 2^{-k}),$$

with $(m_1, \dots, m_N) \in \mathbb{Z}^N$. Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma, k}$$

be the system of dyadic cubes over a fixed chamber Γ , whose construction will be given in Corollary 3.7.

Lemma 2.8. There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \dots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \leq r_B < \frac{2^{-k-1}}{3}$ for some $k \in \mathbb{Z}$, there exists a cube $Q \in \bigcup_{v=1}^{2^N} \mathbb{D}_k^v$ such that

$$B \subset Q \subset 12 \sqrt{N} B.$$

Proof. • **Step 1.** Recall the one-dimensional setting given in [15, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.7). Let $\delta = 1/3$ and define

$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$

where for $k \geq 0$,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I + \delta \mid I \in \mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k < 0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \left\{ \left\lfloor \frac{m}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j}, \frac{m+1}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\rfloor \mid m \in \mathbb{Z} \right\}.$$

These choices together with the nestedness property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k < 0$, k odd.

Fix an interval $J \subset \mathbb{R}$ with $2^{-k-1}/3 \leq |J| < 2^{-k}/3$ for some $k \in \mathbb{Z}$. Now we set $A_k = \{m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for every fixed k and

- (1) $A_k^{(\delta)} = \{\delta + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k \geq 0$,
- (2) $A_k^{(\delta)} = \{\delta + \sum_{j=(k/2)+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k even, and
- (3) $A_k^{(\delta)} = \{\delta + \sum_{j=(k-1)/2+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k odd.

Note that for any two different points $a, b \in A_k \cup A_k^{(\delta)}$, we have $|a - b| \geq 2^{-k}/3 > |J|$. Thus, there is at most one element of $A_k \cup A_k^{(\delta)}$ contained in J . Hence, $J \cap A_k = \emptyset$ or $J \cap A_k^{(\delta)} = \emptyset$. Therefore, J must be contained in some dyadic interval $I \in \mathbb{D}_k^{(0)}$ or $I \in \mathbb{D}_k^{(\delta)}$ and $|I| = 2^{-k} \leq 6|J|$.

• **Step 2.** We extend the construction to $\mathbb{R}^N$ ($N \geq 1$).

For $\vartheta = (\vartheta_1, \vartheta_2, \dots, \vartheta_N) \in \{0, \delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\vartheta)}, \quad \mathbb{D}_k^{(\vartheta)} := \left\{ \prod_{j=1}^N I_j : I_j \in \mathbb{D}_k^{(\vartheta_j)} \right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i -th coordinate projection, we obtain $I_i \in \mathbb{D}_k^{(0)}$ or $I_i \in \mathbb{D}_k^{(\delta)}$ with $\pi_i(B) \subset I_i$ and $|I_i| = 2^{-k} \leq 12r_B$. Let $\vartheta_i \in \{0, \delta\}$ be the corresponding shift so that $I_i \in \mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta := (\vartheta_1, \dots, \vartheta_N)$, and let $Q = \prod_{j=1}^N I_j \in \mathbb{D}_k^{(\vartheta)}$. It follows that $B \subset Q$ and $\ell(Q) = 2^{-k} \leq 12r_B$. Hence, for every $y \in Q$,

$$\|y - x_B\| \leq \sqrt{N}\ell(Q) \leq 12\sqrt{N}r_B.$$

This implies that $Q \subset B(x_B, 12\sqrt{N}r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)} : \vartheta \in \{0, \delta\}^N\}$ as $\mathbb{D}^1, \dots, \mathbb{D}^{2^N}$, we complete the proof of Lemma 2.8. $\square$

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [19] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [4–6, 13, 14, 16, 19, 21, 22]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [13, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Let $\mathcal{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q \in \mathcal{D}}$ is an NWO sequence if

- (1) $\text{supp } e_Q \subseteq cQ$ for some positive constant c independent of Q
- (2) $\|e_Q\|_{L^s(\mathbb{R}^N, dw)} \leq w(Q)^{1/s-1/2}$ for some $s > 2$.

[We need to replace the following Lemma by Hytonen's version on space of homogeneous type. Zhijie Will do it later]

Lemma 2.9. *Let $1 < p < \infty$, $1 \leq q \leq \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. If*

$$T \in S^{p,q}(L^2(\mathbb{R}^N, dw)),$$

then

$$\left\| \left\{ \langle T e_Q, f_Q \rangle_{L^2(\mathbb{R}^N, dw)} \right\}_{Q \in \mathcal{D}} \right\|_{\ell^{p,q}} \leq C_{N,p,q,w} \|T\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))},$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences.

Proof. This property can be found in [19, (1.10)] (see also [13, Section 7]). $\square$

Lemma 2.10. *Let $0 < p < \infty$, $0 < q \leq \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that*

$$A = \sum_{Q \in \mathcal{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q \in \mathcal{D}}$ is a sequence of scalars. We have

$$(2.6) \quad \|A\|_{S^{p,q}} \leq C_{N,p,q,w} \|\{\lambda_Q\}_Q\|_{\ell^{p,q}}.$$

Proof. This property can be found in [19, (1.9)] (see also [13, Section 7]). $\square$

3. ANALYTIC TOOLS FOR WEYL CHAMBER GEOMETRY

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \text{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [9, Definition 2.7] *A connected component of*

$$(3.1) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by

$$R_+ := \{\alpha \in R : \langle x, \alpha \rangle > 0 \text{ for all } x \in \Gamma_+\}, \quad R_- := -R_+.$$

Definition 3.2. [9, Definition 2.9] *A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as*

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \text{with } \lambda_1, \lambda_2 \geq 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. *With the notation given in Definition 3.2, Δ is a basis of W . If we denote by $\Delta = \{\alpha_1, \alpha_2, \dots, \alpha_{N_0}\}$, then the open and closed chambers can be written as*

$$(3.2) \quad \Gamma_+ = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle > 0, i = 1, \dots, N_0\},$$

$$(3.3) \quad \Gamma = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\},$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

such that either $c_i \geq 0$ for all i (when $\alpha \in R_+$), or $c_i \leq 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+ . By [9, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha \rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \text{span}(\Delta)$ and hence,

$$W = \text{span}(R) \subset \text{span}(\Delta).$$

The reverse inclusion $\text{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\text{span}(\Delta) = W$. On the other hand, it follows from [9, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W , and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [9, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (3.2) and (3.3) follow from [9, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+ , note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ , specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. $\square$

Notation 3.4. With the notation given in Lemma 3.3, denote by

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ .

Lemma 3.5. Each Weyl chamber Γ admits the decomposition

$$\Gamma = K(\Gamma) \times W^\perp, \quad K(\Gamma) := \{w \in W : \langle w, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i \rangle = \langle w, \alpha_i \rangle$ for all $i = 1, \dots, N_0$. This, together with (3.3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. $\square$

Corollary 3.6. Let $\Phi_{W^\perp}$ be any linear isomorphism from $W^\perp$ to $\mathbb{R}^{N-N_0}$, and let

$$\begin{aligned} \Phi_W : W &\longrightarrow \mathbb{R}^{N_0}, & \Phi_W(w) &:= (\langle w, \alpha_1 \rangle, \dots, \langle w, \alpha_{N_0} \rangle), \\ \Phi : \mathbb{R}^N = W \oplus W^\perp &\longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, & \Phi(w + z) &:= (\Phi_W(w), \Phi_{W^\perp}(z)). \end{aligned}$$

Then Φ is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W : \langle w, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\}.$$

Since Δ is a basis of W , the Gram matrix $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \rightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^\perp) = \Phi_W(K(\Gamma)) \times \Phi_{W^\perp}(W^\perp) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. $\square$

Corollary 3.7. *Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ .*

Proof. Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}_{\Omega,k}^0$ to be the family of half-open cubes

$$Q = \prod_{i=1}^N I_i, \quad I_i = [m_i 2^{-k}, (m_i + 1) 2^{-k}),$$

where $m_i \in \mathbb{Z}_+$ for $1 \leq i \leq N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \leq N$. Set

$$\mathbb{D}_\Omega^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Omega,k}^0.$$

It is standard that $\mathbb{D}_\Omega^0$ is a system of dyadic cubes over Ω .

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}_{\Omega,k}^0\}, \quad \mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_\Gamma$ is a system of dyadic cubes over Γ . $\square$

Notation 3.8. *Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define*

$$(3.4) \quad \mu(E) := w(\Phi^{-1}(E)).$$

Lemma 3.9. *We have*

$$(3.5) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)}.$$

where for each $v \in R_+$, the coefficients $a_{v,i} \geq 0$ are uniquely determined by the simple-root expansion $v = \sum_{i=1}^{N_0} a_{v,i} \alpha_i$ and satisfy $(a_{v,1}, \dots, a_{v,N_0}) \neq \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω ,

$$\int_{\Omega} f(u, z) d\mu(u, z) = \int_{\Gamma} f(\Phi(x)) dw(x).$$

We make the linear change of variables $(u, z) = \Phi(x)$, so that $x = \Phi^{-1}(u, z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1} du dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i \rangle$ for $i = 1, \dots, N_0$. This implies that

$$(3.6) \quad \langle x, v \rangle = \left\langle x, \sum_{i=1}^{N_0} a_{v,i} \alpha_i \right\rangle = \sum_{i=1}^{N_0} a_{v,i} \langle x, \alpha_i \rangle = \sum_{i=1}^{N_0} a_{v,i} u_i.$$

Substituting these identities into the previous integral, we obtain

$$\begin{aligned} \int_{\Omega} f(u, z) d\mu(u, z) &= \int_{\Gamma} f(\Phi(x)) \left(\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)} \right) dx \\ &= \int_{\Omega} f(u, z) |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)} du dz. \end{aligned}$$

Therefore, (3.5) holds. This completes the proof of Lemma 3.9. $\square$

Lemma 3.10. *Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1, \dots, \alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and*

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \quad k = 1, \dots, M,$$

then the following hold:

- (1) $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$;
- (2) $\Gamma^{(M)} = \Gamma_1$.

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [9, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$ be the simple system associated with Γ_2 . By [9, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1, \dots, i_M \in \{1, \dots, N_0\}$ such that

$$g = \sigma_{\beta_{i_1}} \sigma_{\beta_{i_2}} \cdots \sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \text{Id}, \quad w_k := \sigma_{\beta_{i_1}} \cdots \sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \quad (k = 0, 1, \dots, M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1, \dots, M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

• **Step 1.** Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w \sigma_{\beta} w^{-1} = \sigma_{w(\beta)}, \quad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1} \sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

• **Step 2.** Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \{x \in \mathbb{R}^N : \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta(w_{k-1}(\Gamma_2)) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. $\square$

Definition 3.11. We say that $R' \subset \Omega$ is an axis-parallel rectangle if

$$R' := I_1 \times \dots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

Denote by

$$B_{\Omega}(x_0, r) := \{x \in \Omega : \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 3.12. There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,

$$(3.7) \quad C^{-1} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq \mu(B_{\Omega}(x_0, r)) \leq C r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \quad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant $C_{\text{dbl}, \Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu(B_{\Omega}(x, 2r)) \leq C_{\text{dbl}, \Omega} \mu(B_{\Omega}(x, r)).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $v \in R_+$ define the linear form

$$L_v(u) := a_v \cdot u, \quad a_v := (a_{v,1}, \dots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_v \neq 0$, we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \quad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \leq \|x - x_0\| \leq r.$$

Hence, for every $v \in R_+$,

$$|L_v(u) - L_{v,0}| = |a_v \cdot (u - u_0)| \leq \|a_v\| \cdot \|u - u_0\| \leq L_* r.$$

Since $L_v(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1 (L_{v,0} + r), \quad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0, r)} \phi(u) dx \leq C_2 |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with $C_3 := C_2 \omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \quad \eta := \frac{1}{4(\sqrt{N_0} \lambda + 1)}, \quad \theta := \lambda \eta.$$

These parameters depend only on L_* and m_* , and satisfy

$$\theta > \eta > 0, \quad \sqrt{N_0} \theta + \eta = \eta(\sqrt{N_0} \lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1, \dots, 1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r \mathbf{1}_{N_0}, z_0), \quad B_c := B(x_c, \eta r).$$

We *claim* that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since $\sqrt{N_0} \theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$. This implies that for each $1 \leq i \leq N_0$,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta)r \geq 0,$$

since $u_0 \in \mathbb{R}_+^{N_0}$ and $\theta \geq \eta$. Thus $u \in \mathbb{R}_+^{N_0}$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_v(u)$ on B_c . Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r \mathbf{1}_{N_0} + \Delta_u, \quad z = z_0 + \Delta_z, \quad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular $\|\Delta_u\| \leq \eta r$. Then for all $x = (u, z) \in B_c$ and all $v \in R_+$,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\geq L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\| \|\Delta_u\| \geq L_{v,0} + r K_0 \geq c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \quad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \geq c_0^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu(B_\Omega(x_0, r)) \geq \mu(B_c) = \int_{B_c} \phi(u) dx \geq C_4 |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where $C_5 := C_4 \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \leq C_3 (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \leq C_3 2^{N+\mathcal{K}} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 C_5^{-1} 2^{N+\mathcal{K}} \mu(B_\Omega(x_0, r)),$$

where we set $\mathcal{K} := 2 \sum_{v \in R_+} k(v)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\text{dbl}, \Omega} := C_3 C_5^{-1} 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 3.12. $\square$

Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$(3.8) \quad c_\Phi \|x - y\| \leq \|\Phi(x) - \Phi(y)\| \leq C_\Phi \|x - y\|.$$

Denote by $B_\Gamma(x, r) := \{y \in \Gamma : \|y - x\| < r\}$. It follows from (3.8) that for any $x_0 \in \Gamma$ and $r > 0$,

$$(3.9) \quad B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r).$$

This, together with Lemma 3.12, yields

$$(3.10) \quad \mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Lemma 3.13. *For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis-parallel rectangle R' on Ω such that*

$$(3.11) \quad B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

Furthermore, we have

$$(3.12) \quad \mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis-parallel rectangle

$$(3.13) \quad R' := \left(\prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\}, (u_0)_i + C_\Phi r] \right) \times \left(\prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r, (z_0)_j + C_\Phi r] \right).$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

The second statement follows from the first one together with Lemma 3.12. The proof of Lemma 3.13 is finished. $\square$

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 3.14. *There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,*

$$(3.14) \quad C^{-1} r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)} \leq w(B_\Gamma(x, r)) \leq C r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

In particular, there exists a constant $C_{\text{dbl}, \Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w(B_\Gamma(x, 2r)) \leq C_{\text{dbl}, \Gamma} w(B_\Gamma(x, r)).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$(3.15) \quad w(B_\Gamma(x, r)) \sim w(B(x, r)).$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (3.4) that

$$(3.16) \quad w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))).$$

Applying μ to the inclusions (3.9) and using (3.16) yields

$$(3.17) \quad \mu(B_\Omega(\Phi(x), c_\Phi r)) \leq w(B_\Gamma(x, r)) \leq \mu(B_\Omega(\Phi(x), C_\Phi r)).$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (3.17) with Lemma 3.12 and the equality (3.6), we obtain that

$$(3.18) \quad w(B_\Gamma(x, r)) \simeq r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

This proves (3.14). The second statement follows immediately from (3.14). The third statement follows from the inequalities (3.14) and (2.1). This completes the proof of Theorem 3.14. $\square$

4. PROOF OF UPPER BOUND

Notation 4.1. *Fix a closed Weyl chamber Γ , and choose a linear isomorphism $\Phi : \mathbb{R}^N \longrightarrow \mathbb{R}^N$ as in Corollary 3.6, so that $\Phi(\Gamma) = \Omega$. Let*

$$\mathbb{D}_\Gamma = \bigcup_{j \in \mathbb{Z}} \mathbb{D}_{\Gamma, j}$$

be the dyadic system over Γ associated with this choice of Φ as in Corollary 3.7. [This paragraph of ‘recall’ is just for Dima’s convenience, which can be finally erased]

For each $I \in \mathbb{D}_{\Gamma,j}$, let $Q \in \mathbb{D}_{\Omega,j}^0$ be the unique dyadic cube such that $I = \Phi^{-1}(Q)$. Then $\ell(I) = \ell(Q) = 2^{-j}$. We define

$$I^* := \Phi^{-1}((2Q) \cap \Omega), \quad z_I := \Phi^{-1}(c_Q).$$

Lemma 4.2. Fix $\sigma, \tau \in G$ and $I = \Phi^{-1}(Q) \in \mathbb{D}_{\Gamma}$. There exists $\varepsilon_0 > 0$ independent of σ, τ and I , such that, whenever

$$\text{dist}\left((a, b), \{(a_0, b_0) : a_0 \in [-1/2, 1/2]^N, c_Q + \ell(I)b_0 \in (2Q) \cap \Omega\}\right) < \varepsilon_0$$

and

$$\frac{1}{8} \leq |a - b| \leq \frac{1}{2},$$

the points $X = \sigma\Phi^{-1}(c_Q + \ell(I)a)$ and $Y = \tau\Phi^{-1}(c_Q + \ell(I)b)$ satisfy

$$d(X, Y) \simeq \ell(I), \quad \|X - Y\| \simeq \ell(I) + |\sigma z_I - \tau z_I|, \quad V(X, Y, d(X, Y)) \simeq w(I).$$

Proof. Choose a_0 and b_0 such that $a_0 \in [-1/2, 1/2]^N$, $c_Q + \ell(I)b_0 \in (2Q) \cap \Omega$, and $|a - a_0| + |b - b_0| \lesssim \varepsilon_0$ for some ε_0 to be chosen later. Set $X_0 = \sigma\Phi^{-1}(c_Q + \ell(I)a_0)$ and $Y_0 = \tau\Phi^{-1}(c_Q + \ell(I)b_0)$. If ε_0 is sufficiently small, then $|a_0 - b_0| \gtrsim 1$. Since $c_Q + \ell(I)a_0, c_Q + \ell(I)b_0 \in \Omega$, we apply the G -invariance of d , (2.4), and (3.8) to see that

$$d(X_0, Y_0) = \|\Phi^{-1}(c_Q + \ell(I)a_0) - \Phi^{-1}(c_Q + \ell(I)b_0)\| \gtrsim \ell(I).$$

Since Φ^{-1} is Lipschitz,

$$d(X, Y) \geq d(X_0, Y_0) - \|X - X_0\| - \|Y - Y_0\| \gtrsim \ell(I) - C\varepsilon_0\ell(I) \gtrsim \ell(I)$$

after decreasing ε_0 if necessary. Conversely, by the G -invariance of d and (3.8),

$$d(X, Y) \leq \|\Phi^{-1}(c_Q + \ell(I)a) - \Phi^{-1}(c_Q + \ell(I)b)\| \lesssim \ell(I)|a - b| \lesssim \ell(I).$$

Thus, we have $d(X, Y) \simeq \ell(I)$.

The support condition also gives $|a| + |b| \lesssim 1$, and hence,

$$\|X - \sigma z_I\| + \|Y - \tau z_I\| = \|\Phi^{-1}(c_Q + \ell(I)a) - \Phi^{-1}(c_Q)\| + \|\Phi^{-1}(c_Q + \ell(I)b) - \Phi^{-1}(c_Q)\| \lesssim \ell(I).$$

Therefore

$$(4.19) \quad \|X - Y\| \lesssim \ell(I) + |\sigma z_I - \tau z_I|.$$

On the other hand,

$$\ell(I) \lesssim d(X, Y) \leq \|X - Y\|$$

and

$$|\sigma z_I - \tau z_I| \leq \|X - Y\| + \|X - \sigma z_I\| + \|Y - \tau z_I\| \lesssim \|X - Y\|.$$

Hence,

$$\|X - Y\| \simeq \ell(I) + |\sigma z_I - \tau z_I|,$$

By (3.8), there are fixed constants $c, C > 0$ such that

$$B_{\Gamma}(z_I, c\ell(I)) \subset I \subset B_{\Gamma}(z_I, C\ell(I)).$$

Thus (2.2) and (3.15) imply

$$w(B(z_I, \ell(I))) \simeq w(I).$$

Since $d(X, Y) \simeq \ell(I)$ and $\|X - \sigma z_I\| + \|Y - \tau z_I\| \lesssim \ell(I)$, we apply the doubling property and the G -invariance of dw to see that

$$w(B(X, d(X, Y))) \simeq w(B(\sigma z_I, \ell(I))) = w(B(z_I, \ell(I))) \simeq w(I),$$

and similarly,

$$w(B(Y, d(X, Y))) \simeq w(I).$$

It follows that

$$V(X, Y, d(X, Y)) \simeq w(I).$$

This completes the proof. $\square$

Lemma 4.3. Fix a closed Weyl chamber Γ . Fix $j \in \{1, \dots, N\}$. For $\sigma, \tau \in G$ and $I \in \mathbb{D}_\Gamma$, define

$$(4.20) \quad \theta_{\sigma, \tau, I} = \frac{\ell(I)}{\ell(I) + |\sigma z_I - \tau z_I|}.$$

Then there exist a sequence of real numbers $\{c_{u,v}\}_{u,v \in \mathbb{Z}^N}$ and two families of functions

$$\{e_{u,v,I}^{\sigma, \tau}\}_{u,v \in \mathbb{Z}^N, \sigma, \tau \in G, I \in \mathbb{D}_\Gamma} \quad \text{and} \quad \{h_{u,v,I}^{\sigma, \tau}\}_{u,v \in \mathbb{Z}^N, \sigma, \tau \in G, I \in \mathbb{D}_\Gamma}$$

such that

- (1) $|c_{u,v}| \lesssim_{M_0} (1 + |u| + |v|)^{-M_0}$, for $M_0 > 2N$;
- (2) $\text{supp } e_{u,v,I}^{\sigma, \tau} \subseteq \sigma I$, $\text{supp } h_{u,v,I}^{\sigma, \tau} \subseteq \tau I^*$;
- (3) $|e_{u,v,I}^{\sigma, \tau}| + |h_{u,v,I}^{\sigma, \tau}| \lesssim w(I)^{-1/2}$;
- (4) for almost every (x, y) with $d(x, y) > 0$,

$$(4.21) \quad K_{\text{Dunkl}, j}(x, y) = \sum_{\sigma, \tau \in G} \sum_{u, v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma, \tau, I} e_{u,v,I}^{\sigma, \tau}(x) h_{u,v,I}^{\sigma, \tau}(y).$$

Proof. Fix $\sigma, \tau \in G$. For $x \in \sigma(\Gamma)^\circ$ and $y \in \tau(\Gamma)^\circ$, write $\xi = \Phi(\sigma^{-1}x)$, and $\eta = \Phi(\tau^{-1}y)$. By the G -invariance of the orbit distance, (2.4), and (3.8),

$$d(x, y) = \|\sigma^{-1}x - \tau^{-1}y\| \simeq |\xi - \eta|.$$

Choose $\psi \in C_c^\infty(\mathbb{R}^N)$ such that

$$(4.22) \quad \text{supp } \psi \subset \left\{ z : \frac{1}{8} \leq |z| \leq \frac{1}{2} \right\}, \quad \sum_{m \in \mathbb{Z}} \psi(2^m z) = 1, \quad z \neq 0.$$

If $d(x, y) > 0$, then for each $m \in \mathbb{Z}$, there is a unique $I = \Phi^{-1}(Q) \in \mathbb{D}_{\Gamma, m}$ with $Q \in \mathbb{D}_{\Omega, m}^0$ such that $\sigma^{-1}x \in I$. It therefore follows from (4.22) that

$$K_{\text{Dunkl}, j}(x, y) = \sum_{I \in \mathbb{D}_\Gamma} \chi_{\sigma I}(x) K_{\text{Dunkl}, j}(x, y) \psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right).$$

Moreover, if $x \in \sigma I$ and

$$\psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right) \neq 0,$$

then $|\xi - \eta| \leq \ell(I)/2$. Since $\xi \in Q$ and $\eta \in \Omega$, this implies $\eta \in (2Q) \cap \Omega$, and hence

$$(4.23) \quad y \in \tau I^*.$$

Fix $I = \Phi^{-1}(Q)$ and write $\ell = \ell(I)$. By the construction of $\mathbb{D}_\Omega^0$, we may write

$$Q = \prod_{i=1}^N [m_i \ell, (m_i + 1) \ell),$$

where $m_i \in \mathbb{Z}_+$ for $1 \leq i \leq N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \leq N$. Since the i -th coordinate of c_Q is $(m_i + \frac{1}{2})\ell$, the condition $c_Q + \ell b \in (2Q) \cap \Omega$ gives $-1 \leq b_i < 1$ for every i , with the additional restriction $-\frac{1}{2} \leq b_i < 1$ precisely when $1 \leq i \leq N_0$ and $m_i = 0$. Consequently, the sets

$$\{(a, b) : a \in [-1/2, 1/2]^N, c_Q + \ell b \in (2Q) \cap \Omega\}$$

range over at most 2^{N_0} rectangles. Let ε_0 be as in Lemma 4.2, decreased if necessary so that the ε_0 -neighborhood of each of these rectangles is contained in $(-2, 2)^N \times (-2, 2)^N$. Since there are only finitely many such rectangles, we may choose $\chi_Q \in C_c^\infty((-2, 2)^N \times (-2, 2)^N)$, equal to 1 on the corresponding rectangle and supported in its ε_0 -neighborhood, so that

$$\sup_Q \|\chi_Q\|_{C^L(\mathbb{R}^{2N})} < \infty$$

for every $L \geq 0$.

For $X = \sigma\Phi^{-1}(c_Q + \ell a)$, and $Y = \tau\Phi^{-1}(c_Q + \ell b)$, it follows from Lemmas 4.2 and 2.3 that for every pair of multi-indices α, β ,

$$|\partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(X, Y)| \lesssim \theta_{\sigma,\tau,I} \frac{\ell^{-|\alpha|-|\beta|}}{w(I)}.$$

Define

$$H_{\sigma,\tau,I}(a, b) = \frac{w(I)}{\theta_{\sigma,\tau,I}} \chi_Q(a, b) \psi(a - b) K_{\text{Dunkl},j}(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)).$$

The preceding derivative estimate, the chain rule, and the uniform bounds for χ_Q imply

$$\sup_{\sigma,\tau \in G} \sup_{I \in \mathbb{D}_\Gamma} \|H_{\sigma,\tau,I}\|_{C^L(\mathbb{R}^{2N})} < \infty, \quad L \geq 0.$$

Since $H_{\sigma,\tau,I}$ is compactly supported in $(-2, 2)^N \times (-2, 2)^N$, its 4-periodic extension has the Fourier expansion

$$H_{\sigma,\tau,I}(a, b) = \sum_{u,v \in \mathbb{Z}^N} b_{u,v,I}^{\sigma,\tau} e^{\frac{\pi i}{2} u \cdot a} e^{\frac{\pi i}{2} v \cdot b},$$

and repeated integration by parts yields, for every $L \geq 1$,

$$\sup_{\sigma,\tau \in G} \sup_{I \in \mathbb{D}_\Gamma} |b_{u,v,I}^{\sigma,\tau}| \lesssim_L (1 + |u| + |v|)^{-L}.$$

Choosing an integer $L > M_0$, there is a constant C_{M_0} such that

$$|b_{u,v,I}^{\sigma,\tau}| \leq C_{M_0} (1 + |u| + |v|)^{-M_0} =: c_{u,v}$$

uniformly in σ, τ and I . For $I = \Phi^{-1}(Q)$, define

$$e_{u,v,I}^{\sigma,\tau}(x) = \frac{b_{u,v,I}^{\sigma,\tau}}{c_{u,v}} w(I)^{-1/2} \chi_{\sigma I}(x) \exp\left(\frac{\pi i}{2\ell(I)} u \cdot (\Phi(\sigma^{-1}x) - c_Q)\right),$$

$$h_{u,v,I}^{\sigma,\tau}(y) = w(I)^{-1/2} \chi_{\tau I}(y) \exp\left(\frac{\pi i}{2\ell(I)} v \cdot (\Phi(\tau^{-1}y) - c_Q)\right).$$

Then (2) holds, and

$$|e_{u,v,I}^{\sigma,\tau}| \leq w(I)^{-1/2}, \quad |h_{u,v,I}^{\sigma,\tau}| \leq w(I)^{-1/2},$$

which gives (3).

If $x \in \sigma I$ and $y \in \tau I^*$, set

$$a = \frac{\Phi(\sigma^{-1}x) - c_Q}{\ell(I)}, \quad b = \frac{\Phi(\tau^{-1}y) - c_Q}{\ell(I)}.$$

Since $\sigma^{-1}x \in I = \Phi^{-1}(Q)$, $a \in [-1/2, 1/2]^N$, while

$$c_Q + \ell(I)b = \Phi(\tau^{-1}y) \in (2Q) \cap \Omega$$

because $\tau^{-1}y \in I^*$. Hence $\chi_Q(a, b) = 1$. The Fourier expansion therefore gives

$$K_{\text{Dunkl},j}(x, y) \psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right) = \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \theta_{\sigma,\tau,I} e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y).$$

If $x \in \sigma I$ and $y \notin \tau I^*$, then $h_{u,v,I}^{\sigma,\tau}(y) = 0$. On the other hand, if

$$\psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right) \neq 0,$$

then (4.22) gives $|\xi - \eta| \leq \ell(I)/2$, and hence, as above, $\eta \in (2Q) \cap \Omega$, which would imply $y \in \tau I^*$ by (4.23). Thus the ψ -factor vanishes as well. Hence, for almost every $(x, y) \in \sigma\Gamma \times \tau\Gamma$,

$$\chi_{\sigma I}(x) K_{\text{Dunkl},j}(x, y) \psi \left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)} \right) = \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \theta_{\sigma,\tau,I} e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y).$$

Summing this localized identity over $I \in \mathbb{D}_\Gamma$ and using the dyadic partition of unity, we obtain

$$K_{\text{Dunkl},j}(x, y) = \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma,\tau,I} e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y)$$

for almost every $x \in \sigma(\Gamma)^\circ$, $y \in \tau(\Gamma)^\circ$ with $d(x, y) > 0$.

The interiors of the Weyl chambers cover $\mathbb{R}^N$ up to the reflecting hyperplanes, which have dw -measure zero. Since

$$\text{supp } e_{u,v,I}^{\sigma,\tau} \subset \sigma\Gamma, \quad \text{supp } h_{u,v,I}^{\sigma,\tau} \subset \tau\Gamma,$$

and the interiors of distinct Weyl chambers are disjoint, for almost every $(x, y) \in \mathbb{R}^N \times \mathbb{R}^N$ only the unique pair (σ, τ) satisfying $x \in \sigma(\Gamma)^\circ$ and $y \in \tau(\Gamma)^\circ$ contributes. Thus, summing over $\sigma, \tau \in G$, we obtain (4.21). $\square$

Lemma 4.4. *Fix $s \geq 1$. There exists an exponent $r_s > s$, depending only on s and the structural data of the Dunkl weight, such that for every $I \in \mathbb{D}_\Gamma$ and every $g \in L_{\text{loc}}^1(\mathbb{R}^N)$,*

$$(4.24) \quad \left(\frac{1}{w(I)} \int_I |g - (g)_{I,dx}|^s dw \right)^{1/s} \lesssim \text{osc}_{r_s}(g; I).$$

Moreover, for every $x \in \Gamma$ and $r > 0$,

$$(4.25) \quad \left(\frac{1}{w(B_\Gamma(x, r))} \int_{B_\Gamma(x, r)} |g - (g)_{B_\Gamma(x, r), dx}|^s dw \right)^{1/s} \lesssim \text{osc}_{r_s}(g; B_\Gamma(x, r)).$$

The implicit constants are independent of I , g , x , and r .

Proof. By Lemma 2.1 and the reverse Hölder inequality of A_∞ weights [7, Theorem 7.3.3(c)], there exists $t > 1$ such that

$$\left(\int_B \varpi^t dx \right)^{1/t} \lesssim \int_B \varpi dx$$

for every Euclidean ball B .

Fix $I \in \mathbb{D}_\Gamma$. By the construction of $\mathbb{D}_\Gamma$ and (3.8), there exist constants $c, C > 0$ such that

$$B_\Gamma(z_I, c\ell(I)) \subset I \subset B_\Gamma(z_I, C\ell(I)).$$

For $E = I$, set $B = B(z_I, C\ell(I))$. Then

$$E \subset B, \quad |E| \simeq |B|, \quad w(E) \simeq w(B),$$

where the last equivalence follows from Theorem 3.14 and (3.15).

For $x \in \Gamma$ and $r > 0$, set $E = B_\Gamma(x, r)$ and $B = B(x, r)$. Then

$$E \subset B, \quad |E| \simeq |B|, \quad w(E) \simeq w(B).$$

Consequently, for either of the above choices of E and B ,

$$\left(\frac{1}{|E|} \int_E \varpi^t dx \right)^{1/t} \lesssim \left(\frac{1}{|B|} \int_B \varpi^t dx \right)^{1/t} \lesssim \frac{w(B)}{|B|} \simeq \frac{w(E)}{|E|}.$$

Let $r_s = s'$. Then $r_s > s$. For either choice of E , we apply Hölder's inequality to deduce that

$$\begin{aligned} \left[\frac{1}{w(E)} \int_E |g - (g)_{E,dx}|^s dw \right]^{1/s} &\leq w(E)^{-1/s} \left(\int_E |g - (g)_{E,dx}|^{r_s} dx \right)^{1/r_s} \left(\int_E \varpi^t dx \right)^{1/(st)} \\ &\lesssim \left[\frac{1}{|E|} \int_E |g - (g)_{E,dx}|^{r_s} dx \right]^{1/r_s}. \end{aligned}$$

Taking $E = I$ and $E = B_\Gamma(x, r)$, respectively, we obtain (4.24) and (4.25). This completes the proof of Lemma 4.4. $\square$

Lemma 4.5. *Let $0 < p < \infty$, $0 < q \leq \infty$. Let $\mathcal{A}$ and $\mathcal{B}$ be countable sets, and let $\pi : \mathcal{A} \rightarrow \mathcal{B}$ be a map such that*

$$\sup_{b \in \mathcal{B}} \#\pi^{-1}(\{b\}) \leq M$$

for some $M \geq 1$. Then for every family of nonnegative numbers $\{\lambda_b\}_{b \in \mathcal{B}}$,

$$(4.26) \quad \left\| \{\lambda_{\pi(a)}\}_{a \in \mathcal{A}} \right\|_{\ell^{p,q}(\mathcal{A})} \lesssim M^{1/p} \left\| \{\lambda_b\}_{b \in \mathcal{B}} \right\|_{\ell^{p,q}(\mathcal{B})}.$$

The implicit constant depends only on p and q .

Proof. For $t > 0$, write

$$n_{\mathcal{A}}(t) := \#\{a \in \mathcal{A} : \lambda_{\pi(a)} > t\}, \quad n_{\mathcal{B}}(t) := \#\{b \in \mathcal{B} : \lambda_b > t\}.$$

By the multiplicity assumption,

$$n_{\mathcal{A}}(t) = \sum_{\substack{b \in \mathcal{B} \\ \lambda_b > t}} \#\pi^{-1}(\{b\}) \leq M n_{\mathcal{B}}(t).$$

If $0 < q < \infty$, the distribution-function formula for Lorentz sequence spaces gives

$$\begin{aligned} \|\{\lambda_{\pi(a)}\}_{a \in \mathcal{A}}\|_{\ell^{p,q}}^q &\simeq \int_0^\infty \left[t n_{\mathcal{A}}(t)^{1/p} \right]^q \frac{dt}{t} \\ &\leq M^{q/p} \int_0^\infty \left[t n_{\mathcal{B}}(t)^{1/p} \right]^q \frac{dt}{t} \\ &\lesssim M^{q/p} \|\{\lambda_b\}_{b \in \mathcal{B}}\|_{\ell^{p,q}}^q. \end{aligned}$$

Taking the q -th root proves (4.26). If $q = \infty$, then

$$\|\{\lambda_{\pi(a)}\}\|_{\ell^{p,\infty}} = \sup_{t>0} t n_{\mathcal{A}}(t)^{1/p} \leq M^{1/p} \sup_{t>0} t n_{\mathcal{B}}(t)^{1/p},$$

which proves (4.26). $\square$

Lemma 4.6. *Assume that $N \geq 1$, $1 < p < \infty$ and $1 \leq q \leq \infty$. Fix $s > 2$, $A > 0$ and a closed Weyl chamber Γ . Let r_s be the exponent in Lemma 4.4. For $\tau \in G$ and $f \in OSC^{p,q}(\mathbb{R}^N)$, set*

$$g_\tau(z) := f(\tau z).$$

Then

$$\|\{\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, A\ell(I)))\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)},$$

The implicit constants are independent of f and τ .

Proof. Since τ is orthogonal,

$$\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, A\ell(I))) = \text{osc}_{r_s}(f; \tau B_\Gamma(z_I, A\ell(I))).$$

By Lemma 2.8, there exist $\nu(I) \in \{1, \dots, 2^N\}$ and $J_I \in \mathbb{D}^{\nu(I)}$ such that

$$B(\tau z_I, A\ell(I)) \subset J_I \subset C_N B(\tau z_I, A\ell(I)), \quad \ell(J_I) \simeq \ell(I).$$

Since

$$\tau B_\Gamma(z_I, A\ell(I)) \subset B(\tau z_I, A\ell(I)) \subset J_I$$

and

$$|\tau B_\Gamma(z_I, A\ell(I))| = |B_\Gamma(z_I, A\ell(I))| \simeq \ell(I)^N \simeq |J_I|,$$

we have

$$\text{osc}_{r_s}(f; \tau B_\Gamma(z_I, A\ell(I))) \lesssim \text{osc}_{r_s}(f; J_I).$$

For each $\nu \in \{1, \dots, 2^N\}$, the map

$$\{I \in \mathbb{D}_\Gamma : \nu(I) = \nu\} \longrightarrow \mathbb{D}^\nu, \quad I \longmapsto J_I,$$

has uniformly bounded multiplicity. That is,

$$\#\{I \in \mathbb{D}_\Gamma : \nu(I) = \nu, J_I = J\} \lesssim 1.$$

Therefore, by Lemma 4.5,

$$\|\{\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, A\ell(I)))\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r_s}(f; J_I)\}_{\substack{I \in \mathbb{D}_\Gamma \\ \nu(I)=\nu}} \right\|_{\ell^{p,q}}$$

$$\lesssim \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r_s}(f; J)\}_{J \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} = \|f\|_{OSC^{p,q}(\mathbb{R}^N)}.$$

This completes the proof of Lemma 4.6. $\square$

Lemma 4.7. Assume that $N > 1$. For $\sigma, \tau \in G$, define

$$(4.27) \quad \mathcal{G}_{\sigma,\tau}f(z) := \begin{cases} \frac{|f(\tau z) - f(\sigma z)|}{|\tau z - \sigma z|}, & \tau z \neq \sigma z, \\ 0, & \tau z = \sigma z. \end{cases}$$

Then for every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$,

$$(4.28) \quad \|\mathcal{G}_{\sigma,\tau}f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

The implicit constant is independent of $\sigma, \tau \in G$.

Proof. Let $\mathcal{M}$ be the Hardy–Littlewood maximal operator on $\mathbb{R}^N$. By [2, Theorem 3], after choosing the precise representative of f , there exists a null set $E \subset \mathbb{R}^N$ such that

$$|f(x) - f(y)| \lesssim_N |x - y|(\mathcal{M}(|\nabla f|)(x) + \mathcal{M}(|\nabla f|)(y)), \quad x, y \in \mathbb{R}^N \setminus E.$$

Since σ and τ are orthogonal, they preserve Lebesgue measure. Hence, for almost every $z \in \mathbb{R}^N$ with $\tau z \neq \sigma z$,

$$\mathcal{G}_{\sigma,\tau}f(z) \lesssim_N \mathcal{M}(|\nabla f|)(\tau z) + \mathcal{M}(|\nabla f|)(\sigma z).$$

The same estimate holds when $\tau z = \sigma z$ by the definition of $\mathcal{G}_{\sigma,\tau}f$. Therefore, using again the measure-preserving property of σ and τ and the L^N -boundedness of $\mathcal{M}$, which is valid since $N > 1$,

$$\begin{aligned} \|\mathcal{G}_{\sigma,\tau}f\|_{L^N(\mathbb{R}^N)} &\lesssim_N \|\mathcal{M}(|\nabla f|) \circ \tau\|_{L^N} + \|\mathcal{M}(|\nabla f|) \circ \sigma\|_{L^N} \\ &= 2\|\mathcal{M}(|\nabla f|)\|_{L^N} \lesssim_N \|\nabla f\|_{L^N(\mathbb{R}^N)}. \end{aligned}$$

The implicit constant is independent of $\sigma, \tau \in G$. $\square$

Lemma 4.8. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. For $\sigma, \tau \in G$ and $I \in \mathbb{D}_\Gamma$, let $\theta_{\sigma,\tau,I}$ be as in (4.20). For $f \in L^1_{\text{loc}}(\mathbb{R}^N)$, define

$$(4.29) \quad a_{\sigma,\tau,I}(f) := \frac{1}{|I|} \int_I (f(\tau z) - f(\sigma z)) dz.$$

Then for every $f \in OSC^{p,q}(\mathbb{R}^N)$,

$$(4.30) \quad \left\| \{\theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)}.$$

The implicit constant is independent of $\sigma, \tau \in G$.

Proof. The proof is divided into three cases.

Case 1. Suppose that $N \geq 1$, $N < p < \infty$, and $1 \leq q \leq \infty$.

Fix $\sigma, \tau \in G$, and choose $r > Np/(p - N)$ so that

$$1 - \frac{N}{r} - \frac{N}{p} > 0.$$

For $k \geq 0$, let

$$\mathcal{I}_k := \{I \in \mathbb{D}_\Gamma : 2^{-k-1} < \theta_{\sigma,\tau,I} \leq 2^{-k}\}.$$

Then $\mathbb{D}_\Gamma = \bigcup_{k \geq 0} \mathcal{I}_k$ disjointly.

By the construction of $\mathbb{D}_\Gamma$ and (3.8),

$$|I| \simeq \ell(I)^N, \quad |z - z_I| \lesssim \ell(I), \quad z \in I.$$

Hence, for $I \in \mathcal{I}_k$,

$$\sigma I \cup \tau I \subset B_I := B(\sigma z_I, C2^k \ell(I))$$

with a structural constant C , and $|B_I| \simeq 2^{kN}|I|$. Therefore, by the orthogonality of σ, τ and Hölder's inequality,

$$\begin{aligned} |a_{\sigma,\tau,I}(f)| &\leq \frac{1}{|I|} \int_{\tau I} |f - (f)_{B_I,dx}| dx + \frac{1}{|I|} \int_{\sigma I} |f - (f)_{B_I,dx}| dx \\ &\lesssim \left(\frac{|B_I|}{|I|} \right)^{1/r} \left(\frac{1}{|B_I|} \int_{B_I} |f - (f)_{B_I,dx}|^r dx \right)^{1/r}. \end{aligned}$$

Using $|B_I|/|I| \simeq 2^{kN}$, we obtain

$$(4.31) \quad \theta_{\sigma,\tau,I} |a_{\sigma,\tau,I}(f)| \lesssim 2^{-k(1-N/r)} \text{osc}_r(f; B_I), \quad I \in \mathcal{I}_k.$$

By Lemma 2.8, for every $I \in \mathcal{I}_k$ there are $\nu(I) \in \{1, \dots, 2^N\}$ and $J_I \in \mathbb{D}^{\nu(I)}$ such that

$$B_I \subset J_I \subset C_N B_I.$$

Consequently,

$$\ell(J_I) \simeq 2^k \ell(I), \quad |J_I| \simeq |B_I|, \quad \text{osc}_r(f; B_I) \lesssim \text{osc}_r(f; J_I).$$

For fixed k and ν , we claim that

$$\sup_{J \in \mathbb{D}^\nu} \# \{I \in \mathcal{I}_k : \nu(I) = \nu, J_I = J\} \lesssim 2^{kN}.$$

Indeed, if $J_I = J$, then $\ell(I) \simeq 2^{-k} \ell(J)$. Since both I and J have dyadic side-lengths, only finitely many generations of I , uniformly in k and J , can occur. For each such generation, $B_I \subset J$ implies $\sigma z_I \in J$. If $I = \Phi^{-1}(Q)$ and $I' = \Phi^{-1}(Q')$ are distinct cubes of the same generation, then

$$|c_Q - c_{Q'}| \geq \ell(I).$$

Since $c_Q = \Phi(z_I)$, $c_{Q'} = \Phi(z_{I'})$, and σ is orthogonal, (3.8) gives

$$|\sigma z_I - \sigma z_{I'}| = |z_I - z_{I'}| \geq C_\Phi^{-1} |c_Q - c_{Q'}| \geq C_\Phi^{-1} \ell(I).$$

Thus, for each fixed generation, the balls

$$B\left(\sigma z_I, \frac{C_\Phi^{-1} \ell(I)}{2}\right), \quad I \in \mathcal{I}_k, \quad \nu(I) = \nu, \quad J_I = J, \quad \ell(I) = 2^{-j},$$

are pairwise disjoint. Since their centers belong to J and $\ell(I) \lesssim \ell(J)$, all these balls are contained in a fixed dilation CJ of J . Hence,

$$\# \{I \in \mathcal{I}_k : \nu(I) = \nu, J_I = J, \ell(I) = 2^{-j}\} \ell(I)^N \lesssim |CJ| \lesssim \ell(J)^N,$$

and therefore,

$$\#\{I \in \mathcal{I}_k : \nu(I) = \nu, J_I = J, \ell(I) = 2^{-j}\} \lesssim \left(\frac{\ell(J)}{\ell(I)}\right)^N \lesssim 2^{kN}.$$

Summing over the finitely many possible generations yields

$$\#\{I \in \mathcal{I}_k : \nu(I) = \nu, J_I = J\} \lesssim 2^{kN}.$$

Thus, by Lemma 4.5 and (4.31),

$$\begin{aligned} \|\{\theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f)\}_{I \in \mathcal{I}_k}\|_{\ell^{p,q}} &\lesssim 2^{-k(1-N/r)} \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_r(f; J_I)\}_{\substack{I \in \mathcal{I}_k \\ \nu(I)=\nu}} \right\|_{\ell^{p,q}} \\ &\lesssim 2^{-k(1-N/r-N/p)} \sum_{\nu=1}^{2^N} \|\{\text{osc}_r(f; J)\}_{J \in \mathbb{D}^\nu}\|_{\ell^{p,q}} \\ &\lesssim 2^{-k(1-N/r-N/p)} \|f\|_{OS C^{p,q}(\mathbb{R}^N)}. \end{aligned}$$

Therefore,

$$\begin{aligned} \|\{\theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} &\lesssim \sum_{k=0}^{\infty} \|\{\theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f)\}_{I \in \mathcal{I}_k}\|_{\ell^{p,q}} \\ &\lesssim \sum_{k=0}^{\infty} 2^{-k(1-N/r-N/p)} \|f\|_{OS C^{p,q}(\mathbb{R}^N)} \\ &\lesssim \|f\|_{OS C^{p,q}(\mathbb{R}^N)}. \end{aligned}$$

Case 2. Suppose that $N > 1$, $p = N$ and $q = \infty$.

By part 1 of Lemma 6.2,

$$\|\nabla f\|_{L^N(\mathbb{R}^N)} \simeq \|f\|_{OS C^{N,\infty}(\mathbb{R}^N)}.$$

For $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$ and $z \in I$, we use the bi-Lipschitz property (3.8) to see that

$$|z - z_I| \lesssim |\Phi(z) - \Phi(z_I)| \leq \frac{\sqrt{N}}{2} \ell(Q) \lesssim \ell(I).$$

Since σ and τ are orthogonal,

$$|\tau z - \sigma z| \leq 2|z - z_I| + |\tau z_I - \sigma z_I| \lesssim \ell(I) + |\tau z_I - \sigma z_I|.$$

Hence, by the definition of $\mathcal{G}_{\sigma,\tau} f$ (see (4.27)) and (4.20),

$$\begin{aligned} \theta_{\sigma,\tau,I} |a_{\sigma,\tau,I}(f)| &\leq \frac{\theta_{\sigma,\tau,I}}{|I|} \int_I |f(\tau z) - f(\sigma z)| dz \\ &\lesssim \frac{\ell(I)}{\ell(I) + |\sigma z_I - \tau z_I|} (\ell(I) + |\tau z_I - \sigma z_I|) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma,\tau} f(z) dz \\ &\lesssim \ell(I) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma,\tau} f(z) dz. \end{aligned}$$

Set

$$F_{\sigma,\tau}(u) := \mathcal{G}_{\sigma,\tau} f(\Phi^{-1}u) \chi_\Omega(u).$$

By the change of variables $u = \Phi(z)$, together with Lemma 4.7,

$$\|F_{\sigma,\tau}\|_{L^N(\mathbb{R}^N)} \lesssim \|\mathcal{G}_{\sigma,\tau} f\|_{L^N(\mathbb{R}^N)} \lesssim \|\nabla f\|_{L^N(\mathbb{R}^N)},$$

uniformly in $\sigma, \tau \in G$. Moreover, for $I = \Phi^{-1}(Q)$,

$$\ell(I) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma, \tau} f(z) dz = \ell(Q) \frac{1}{|Q|} \int_Q F_{\sigma, \tau}(u) du = |Q|^{\frac{1}{N}-1} \int_Q F_{\sigma, \tau}(u) du.$$

By the construction of $\mathbb{D}_\Gamma$, every $I \in \mathbb{D}_\Gamma$ is of the form $I = \Phi^{-1}(Q)$ for a unique $Q \in \mathbb{D}_\Omega^0 \subset \mathbb{D}^0$. Applying Lemma 6.1 with $s = N$, we deduce that

$$\left\| \left\{ \ell(I) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma, \tau} f(z) dz \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N, \infty}} \leq \left\| \left\{ |Q|^{\frac{1}{N}-1} \int_Q F_{\sigma, \tau}(u) du \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{N, \infty}} \lesssim \|F_{\sigma, \tau}\|_{L^N(\mathbb{R}^N)}.$$

Combining the preceding estimates, we deduce that

$$\left\| \{\theta_{\sigma, \tau, I} a_{\sigma, \tau, I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N, \infty}} \lesssim \|\nabla f\|_{L^N(\mathbb{R}^N)} \simeq \|f\|_{OS C^{N, \infty}(\mathbb{R}^N)},$$

where the final equivalence follows from Lemma 6.2, and where the implicit constant is uniform in $\sigma, \tau \in G$.

Case 3. Suppose that $N > 1$ and either $1 < p < N$, $1 \leq q \leq \infty$, or $p = N$, $1 \leq q < \infty$.

By parts 2 and 3 of Lemma 6.2, f is constant almost everywhere. Hence,

$$f(\tau z) - f(\sigma z) = 0 \quad \text{for almost every } z \in \mathbb{R}^N.$$

Hence, by (4.29), $a_{\sigma, \tau, I}(f) = 0$ for every $I \in \mathbb{D}_\Gamma$. Therefore,

$$\left\| \{\theta_{\sigma, \tau, I} a_{\sigma, \tau, I}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} = 0.$$

This completes the proof of Lemma 4.8. $\square$

Lemma 4.9. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. Fix $s > 2$ and a closed Weyl chamber Γ . Let r_s be the exponent in Lemma 4.4. For $f \in L^1_{\text{loc}}(\mathbb{R}^N)$ and $\sigma, \tau \in G$, define

$$(4.32) \quad H_{\sigma, \tau}(f)(z) := f(\tau z) - f(\sigma z), \quad z \in \mathbb{R}^N.$$

Then for any $f \in OSC^{p, q}(\mathbb{R}^N)$,

$$\left\| \{\text{osc}_{r_s}(H_{\sigma, \tau}(f); I)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} \lesssim \|f\|_{OSC^{p, q}(\mathbb{R}^N)},$$

The implicit constant is independent of $\sigma, \tau \in G$.

Proof. Since σ and τ are orthogonal,

$$(H_{\sigma, \tau}(f))_{I, dx} = (f)_{\tau I, dx} - (f)_{\sigma I, dx},$$

and hence, by Minkowski's inequality,

$$\text{osc}_{r_s}(H_{\sigma, \tau}(f); I) \leq \text{osc}_{r_s}(f; \tau I) + \text{osc}_{r_s}(f; \sigma I).$$

Fix $\gamma \in \{\sigma, \tau\}$. By the construction of $\mathbb{D}_\Gamma$ and (3.8),

$$\gamma I \subset B(\gamma z_I, C\ell(I)), \quad |\gamma I| = |I| \simeq \ell(I)^N.$$

By Lemma 2.8, for every $I \in \mathbb{D}_\Gamma$ there exist $\nu_\gamma(I) \in \{1, \dots, 2^N\}$ and $J_I^\gamma \in \mathbb{D}^{\nu_\gamma(I)}$ such that

$$\gamma I \subset B(\gamma z_I, C\ell(I)) \subset J_I^\gamma \subset C_N B(\gamma z_I, C\ell(I)).$$

Thus, $\ell(J_I^\gamma) \simeq \ell(I)$, and $|J_I^\gamma| \simeq |\gamma I|$. This implies that

$$\text{osc}_{r_s}(f; \gamma I) \lesssim \text{osc}_{r_s}(f; J_I^\gamma).$$

Note that for each $\nu \in \{1, \dots, 2^N\}$, the map

$$I \mapsto J_I^\gamma, \quad \{I \in \mathbb{D}_\Gamma : \nu_\gamma(I) = \nu\} \longrightarrow \mathbb{D}^\nu,$$

has uniformly bounded multiplicity. That is,

$$\sup_{J \in \mathbb{D}^\nu} \# \{I \in \mathbb{D}_\Gamma : \nu_\gamma(I) = \nu, J_I^\gamma = J\} \lesssim 1.$$

Therefore, by Lemma 4.5,

$$\begin{aligned} \left\| \{\text{osc}_{r_s}(f; \gamma I)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} &\lesssim \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r_s}(f; J_I^\gamma)\}_{\substack{I \in \mathbb{D}_\Gamma \\ \nu_\gamma(I)=\nu}} \right\|_{\ell^{p,q}} \\ &\lesssim \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r_s}(f; J)\}_{J \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} = \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}. \end{aligned}$$

Applying this estimate with $\gamma = \sigma$ and $\gamma = \tau$, we obtain

$$\left\| \{\text{osc}_{r_s}(H_{\sigma,\tau}(f); I)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} \lesssim \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}.$$

This completes the proof of Lemma 4.9. $\square$

Lemma 4.10. *Assume that $N \geq 1$, $1 < p < \infty$ and $1 \leq q \leq \infty$. Fix $s > 2$, $j \in \{1, \dots, N\}$, and a closed Weyl chamber Γ . For $f \in \text{OSC}^{p,q}(\mathbb{R}^N)$, the off-diagonal kernel of $[R_{\text{Dunkl},j}, M_f]$, denoted by $K_{[R_{\text{Dunkl},j}, M_f]}(x, y)$, admits the following factorization: for almost every (x, y) with $d(x, y) > 0$,*

$$(4.33) \quad K_{[R_{\text{Dunkl},j}, M_f]}(x, y) = \sum_{\varepsilon=0}^3 \sum_{\sigma, \tau \in G} \sum_{u, v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma, \tau, I, \varepsilon}(f) E_{u,v,I,\varepsilon}^{\sigma, \tau}(x) H_{u,v,I,\varepsilon}^{\sigma, \tau}(y),$$

where the real numbers $\{c_{u,v}\}_{u,v \in \mathbb{Z}^N}$, coefficients $\{A_{\sigma, \tau, I, \varepsilon}(f)\}_{\sigma, \tau \in G, I \in \mathbb{D}_\Gamma, 0 \leq \varepsilon \leq 3}$, and functions $E_{u,v,I,\varepsilon}^{\sigma, \tau}, H_{u,v,I,\varepsilon}^{\sigma, \tau}$ satisfy

(1) for every $M_0 > 2N$,

$$|c_{u,v}| \lesssim_{M_0} (1 + |u| + |v|)^{-M_0};$$

(2) for every $\sigma, \tau \in G$, $u, v \in \mathbb{Z}^N$, $I \in \mathbb{D}_\Gamma$, and $0 \leq \varepsilon \leq 3$,

$$\text{supp } E_{u,v,I,\varepsilon}^{\sigma, \tau} \subseteq \sigma I, \quad \text{supp } H_{u,v,I,\varepsilon}^{\sigma, \tau} \subseteq \tau I^*;$$

(3) There exists a constant $C > 0$, independent of $\sigma, \tau, u, v, I, \varepsilon$, such that

$$\|E_{u,v,I,\varepsilon}^{\sigma, \tau}\|_{L^s(dw)} + \|H_{u,v,I,\varepsilon}^{\sigma, \tau}\|_{L^s(dw)} \leq C w(I)^{\frac{1}{s} - \frac{1}{2}};$$

(4) there exists a constant $C > 0$, independent of $\sigma, \tau, \varepsilon$, such that

$$\left\| \{A_{\sigma, \tau, I, \varepsilon}(f)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} \leq C \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}.$$

Proof. Fix $s > 2$, and let $r_s > s$ be the exponent in Lemma 4.4. By the definition of $\text{OSC}^{p,q}(\mathbb{R}^N)$ and the independence of the defining exponent,

$$f \in L_{\text{loc}}^{r_s}(\mathbb{R}^N, dx) \subset L_{\text{loc}}^s(\mathbb{R}^N, dw).$$

Step 1. The kernel factorization (4.33).

By Lemma 4.3, for almost every (x, y) with $d(x, y) > 0$,

$$K_{\text{Dunkl},j}(x, y) = \sum_{\sigma, \tau \in G} \sum_{u, v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma, \tau, I} e_{u,v,I}^{\sigma, \tau}(x) h_{u,v,I}^{\sigma, \tau}(y),$$

with $c_{u,v}$, $e_{u,v,I}^{\sigma, \tau}$, and $h_{u,v,I}^{\sigma, \tau}$ satisfying the corresponding properties stated in Lemma 4.3. In particular,

$$|c_{u,v}| \lesssim_{M_0} (1 + |u| + |v|)^{-M_0}, \quad M_0 > 2N.$$

Fix $\sigma, \tau \in G$, $u, v \in \mathbb{Z}^N$, and $I \in \mathbb{D}_\Gamma$. On the support of $e_{u,v,I}^{\sigma, \tau}(x) h_{u,v,I}^{\sigma, \tau}(y)$, we may write $x = \sigma\xi$, $y = \tau\eta$, where $\xi \in I$ and $\eta \in I^*$. Then we decompose $f(y) - f(x)$ into the sum of a same-chamber term and a cross-chamber term:

$$(4.34) \quad f(y) - f(x) = (f(\tau\eta) - f(\tau\xi)) + (f(\tau\xi) - f(\sigma\xi)).$$

Set $g_\tau(z) := f(\tau z)$. Recall that c_Φ is the constant given in (3.8), and let $C_1 := \frac{2\sqrt{N}}{c_\Phi}$. Define

$$A_{\sigma, \tau, I, 0}(f) := A_{\sigma, \tau, I, 1}(f) := \theta_{\sigma, \tau, I} \text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))).$$

If $\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) > 0$, define

$$E_{u,v,I,0}^{\sigma, \tau}(x) := e_{u,v,I}^{\sigma, \tau}(x),$$

$$H_{u,v,I,0}^{\sigma, \tau}(y) := (\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} (f(y) - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}) h_{u,v,I}^{\sigma, \tau}(y),$$

and

$$E_{u,v,I,1}^{\sigma, \tau}(x) := -(\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} (f(\tau\sigma^{-1}x) - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}) e_{u,v,I}^{\sigma, \tau}(x),$$

$$H_{u,v,I,1}^{\sigma, \tau}(y) := h_{u,v,I}^{\sigma, \tau}(y).$$

If $\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) = 0$, set these four functions equal to zero.

Suppose first that $\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) > 0$. Since $y = \tau\eta$ and $\tau\sigma^{-1}x = \tau\xi$, we have

$$\begin{aligned} & \theta_{\sigma, \tau, I} (f(\tau\eta) - f(\tau\xi)) e_{u,v,I}^{\sigma, \tau}(x) h_{u,v,I}^{\sigma, \tau}(y) \\ &= \theta_{\sigma, \tau, I} \left((f(y) - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}) - (f(\tau\sigma^{-1}x) - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}) \right) e_{u,v,I}^{\sigma, \tau}(x) h_{u,v,I}^{\sigma, \tau}(y) \\ &= \theta_{\sigma, \tau, I} \text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) \times \\ & \quad \times \left(e_{u,v,I}^{\sigma, \tau}(x) (\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} (f(y) - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}) h_{u,v,I}^{\sigma, \tau}(y) + \right. \\ & \quad \left. + \left(-(\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} (f(\tau\sigma^{-1}x) - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}) e_{u,v,I}^{\sigma, \tau}(x) \right) h_{u,v,I}^{\sigma, \tau}(y) \right) \\ &= \theta_{\sigma, \tau, I} \text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) \sum_{\varepsilon=0}^1 E_{u,v,I,\varepsilon}^{\sigma, \tau}(x) H_{u,v,I,\varepsilon}^{\sigma, \tau}(y) \\ (4.35) \quad &= \sum_{\varepsilon=0}^1 A_{\sigma, \tau, I, \varepsilon}(f) E_{u,v,I,\varepsilon}^{\sigma, \tau}(x) H_{u,v,I,\varepsilon}^{\sigma, \tau}(y). \end{aligned}$$

If $\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) = 0$, then by the definition of osc_{r_s} ,

$$g_\tau = (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx} \quad dx\text{-a.e. on } B_\Gamma(z_I, C_1 \ell(I)),$$

and hence also

$$g_\tau = (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx} \quad dw\text{-a.e. on } B_\Gamma(z_I, C_1 \ell(I)).$$

Since $I \subset I^* \subset B_\Gamma(z_I, C_1 \ell(I))$, we have $f(\tau\eta) - f(\tau\xi) = 0$ for almost every $\xi \in I$ and $\eta \in I^*$. By the definitions above, the last expression in (4.35) also vanishes. Hence, in either case we have

$$\theta_{\sigma,\tau,I}(f(\tau\eta) - f(\tau\xi))e_{u,v,I}^{\sigma,\tau}(x)h_{u,v,I}^{\sigma,\tau}(y) = \sum_{\varepsilon=0}^1 A_{\sigma,\tau,I,\varepsilon}(f)E_{u,v,I,\varepsilon}^{\sigma,\tau}(x)H_{u,v,I,\varepsilon}^{\sigma,\tau}(y).$$

We next turn to the cross-chamber term $f(\tau\xi) - f(\sigma\xi)$. Let $a_{\sigma,\tau,I}(f)$ be as in (4.29), and let $H_{\sigma,\tau}(f)$ be as in (4.32). By the definition of $a_{\sigma,\tau,I}(f)$, we have

$$a_{\sigma,\tau,I}(f) = (H_{\sigma,\tau}(f))_{I,dx}.$$

Hence, by (4.24),

$$\left(\frac{1}{w(I)} \int_I |H_{\sigma,\tau}(f) - a_{\sigma,\tau,I}(f)|^s dw \right)^{1/s} \lesssim \text{osc}_{r_s}(H_{\sigma,\tau}(f); I).$$

Define

$$\begin{aligned} A_{\sigma,\tau,I,2}(f) &:= \theta_{\sigma,\tau,I} \text{osc}_{r_s}(H_{\sigma,\tau}(f); I), & A_{\sigma,\tau,I,3}(f) &:= \theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f), & H_{u,v,I,2}^{\sigma,\tau}(y) &:= h_{u,v,I}^{\sigma,\tau}(y), \\ E_{u,v,I,2}^{\sigma,\tau}(x) &:= \begin{cases} (\text{osc}_{r_s}(H_{\sigma,\tau}(f); I))^{-1} (H_{\sigma,\tau}(f)(\sigma^{-1}x) - a_{\sigma,\tau,I}(f)) e_{u,v,I}^{\sigma,\tau}(x), & \text{osc}_{r_s}(H_{\sigma,\tau}(f); I) > 0, \\ 0, & \text{osc}_{r_s}(H_{\sigma,\tau}(f); I) = 0, \end{cases} \end{aligned}$$

and

$$E_{u,v,I,3}^{\sigma,\tau}(x) := e_{u,v,I}^{\sigma,\tau}(x), \quad H_{u,v,I,3}^{\sigma,\tau}(y) := h_{u,v,I}^{\sigma,\tau}(y).$$

Suppose first that $\text{osc}_{r_s}(H_{\sigma,\tau}(f); I) > 0$. Since $\sigma^{-1}x = \xi$, we have

$$\begin{aligned} & \theta_{\sigma,\tau,I}(f(\tau\xi) - f(\sigma\xi))e_{u,v,I}^{\sigma,\tau}(x)h_{u,v,I}^{\sigma,\tau}(y) \\ &= \theta_{\sigma,\tau,I}H_{\sigma,\tau}(f)(\sigma^{-1}x)e_{u,v,I}^{\sigma,\tau}(x)h_{u,v,I}^{\sigma,\tau}(y) \\ &= \theta_{\sigma,\tau,I}(H_{\sigma,\tau}(f)(\sigma^{-1}x) - a_{\sigma,\tau,I}(f) + a_{\sigma,\tau,I}(f))e_{u,v,I}^{\sigma,\tau}(x)h_{u,v,I}^{\sigma,\tau}(y) \\ &= \theta_{\sigma,\tau,I} \text{osc}_{r_s}(H_{\sigma,\tau}(f); I) \left((\text{osc}_{r_s}(H_{\sigma,\tau}(f); I))^{-1} (H_{\sigma,\tau}(f)(\sigma^{-1}x) - a_{\sigma,\tau,I}(f)) e_{u,v,I}^{\sigma,\tau}(x) \right) h_{u,v,I}^{\sigma,\tau}(y) \\ &\quad + \theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) \\ &= \theta_{\sigma,\tau,I} \text{osc}_{r_s}(H_{\sigma,\tau}(f); I) E_{u,v,I,2}^{\sigma,\tau}(x) H_{u,v,I,2}^{\sigma,\tau}(y) + \theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) \\ (4.36) \quad &= \sum_{\varepsilon=2}^3 A_{\sigma,\tau,I,\varepsilon}(f) E_{u,v,I,\varepsilon}^{\sigma,\tau}(x) H_{u,v,I,\varepsilon}^{\sigma,\tau}(y). \end{aligned}$$

If $\text{osc}_{r_s}(H_{\sigma,\tau}(f); I) = 0$, then, since $a_{\sigma,\tau,I}(f) = (H_{\sigma,\tau}(f))_{I,dx}$, the definition of $\text{osc}_{r_s}(H_{\sigma,\tau}(f); I)$ gives

$$H_{\sigma,\tau}(f) = a_{\sigma,\tau,I}(f) \quad dx\text{-a.e. on } I,$$

and hence also

$$H_{\sigma,\tau}(f) = a_{\sigma,\tau,I}(f) \quad dw\text{-a.e. on } I.$$

Since $\xi \in I$, we have

$$f(\tau\xi) - f(\sigma\xi) = a_{\sigma,\tau,I}(f) \quad dw\text{-a.e. on } I.$$

Moreover,

$$A_{\sigma,\tau,I,2}(f) = \theta_{\sigma,\tau,I} \text{osc}_{r_s}(H_{\sigma,\tau}(f); I) = 0,$$

and hence, the last expression in (4.36) reduces to $\theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y)$. Hence, in either case we have

$$\theta_{\sigma,\tau,I}(f(\tau\xi) - f(\sigma\xi)) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) = \sum_{\varepsilon=2}^3 A_{\sigma,\tau,I,\varepsilon}(f) E_{u,v,I,\varepsilon}^{\sigma,\tau}(x) H_{u,v,I,\varepsilon}^{\sigma,\tau}(y).$$

Combining (4.35) and (4.36), and using (4.34), we obtain

$$\begin{aligned} & \theta_{\sigma,\tau,I}(f(y) - f(x)) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) \\ &= \theta_{\sigma,\tau,I}(f(\tau\eta) - f(\tau\xi) + f(\tau\xi) - f(\sigma\xi)) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) \\ &= \theta_{\sigma,\tau,I} \text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) \sum_{\varepsilon=0}^1 E_{u,v,I,\varepsilon}^{\sigma,\tau}(x) H_{u,v,I,\varepsilon}^{\sigma,\tau}(y) \\ & \quad + \theta_{\sigma,\tau,I} \text{osc}_{r_s}(H_{\sigma,\tau}(f); I) E_{u,v,I,2}^{\sigma,\tau}(x) H_{u,v,I,2}^{\sigma,\tau}(y) \\ & \quad + \theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) \\ (4.37) \quad &= \sum_{\varepsilon=0}^3 A_{\sigma,\tau,I,\varepsilon}(f) E_{u,v,I,\varepsilon}^{\sigma,\tau}(x) H_{u,v,I,\varepsilon}^{\sigma,\tau}(y). \end{aligned}$$

Since

$$K_{[R_{\text{Dunkl},j}, M_f]}(x, y) = (f(y) - f(x)) K_{\text{Dunkl},j}(x, y), \quad d(x, y) > 0,$$

we apply Lemma 4.3 and (4.37) to see that for almost every (x, y) with $d(x, y) > 0$,

$$\begin{aligned} K_{[R_{\text{Dunkl},j}, M_f]}(x, y) &= \sum_{\sigma,\tau \in G} \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma,\tau,I}(f(y) - f(x)) e_{u,v,I}^{\sigma,\tau}(x) h_{u,v,I}^{\sigma,\tau}(y) \\ &= \sum_{\varepsilon=0}^3 \sum_{\sigma,\tau \in G} \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) E_{u,v,I,\varepsilon}^{\sigma,\tau}(x) H_{u,v,I,\varepsilon}^{\sigma,\tau}(y), \end{aligned}$$

which proves (4.33).

Step 2. Verification of (2) and (3).

By Lemma 4.3, for every $\varepsilon = 0, 1, 2, 3$,

$$\text{supp } E_{u,v,I,\varepsilon}^{\sigma,\tau} \subseteq \sigma I, \quad \text{supp } H_{u,v,I,\varepsilon}^{\sigma,\tau} \subseteq \tau I^*.$$

We next verify the required $L^s(dw)$ estimates. For $\varepsilon = 0, 1$, suppose first that

$$\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) > 0.$$

Using Lemma 4.3, the G -invariance of dw , and $I \subset I^* \subset B_\Gamma(z_I, C_1 \ell(I))$, we obtain

$$\begin{aligned} \|H_{u,v,I,0}^{\sigma,\tau}\|_{L^s(dw)} &\lesssim (\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} w(I)^{-1/2} \left(\int_{I^*} |g_\tau - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}|^s dw \right)^{1/s} \\ &\leq (\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} w(I)^{-1/2} \left(\int_{B_\Gamma(z_I, C_1 \ell(I))} |g_\tau - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}|^s dw \right)^{1/s} \\ &\lesssim w(I)^{-1/2} w(B_\Gamma(z_I, C_1 \ell(I)))^{1/s} \lesssim w(I)^{\frac{1}{s} - \frac{1}{2}}, \\ \|E_{u,v,I,1}^{\sigma,\tau}\|_{L^s(dw)} &\lesssim (\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} w(I)^{-1/2} \left(\int_I |g_\tau - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I)), dx}|^s dw \right)^{1/s} \end{aligned}$$

$$\begin{aligned} &\leq (\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))))^{-1} w(I)^{-1/2} \left(\int_{B_\Gamma(z_I, C_1 \ell(I))} |g_\tau - (g_\tau)_{B_\Gamma(z_I, C_1 \ell(I), dx)}|^s dw \right)^{1/s} \\ &\lesssim w(I)^{-1/2} w(B_\Gamma(z_I, C_1 \ell(I)))^{1/s} \lesssim w(I)^{\frac{1}{s}-\frac{1}{2}}, \end{aligned}$$

where we used (4.25) in the penultimate inequality. If $\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I))) = 0$, the same estimates are immediate from $H_{u,v,I,0}^{\sigma,\tau} = E_{u,v,I,1}^{\sigma,\tau} = 0$. Moreover, since

$$w(I^*) \leq w(B_\Gamma(z_I, C_1 \ell(I))) \lesssim w(I),$$

we apply Lemma 4.3 to see that

$$\|E_{u,v,I,0}^{\sigma,\tau}\|_{L^s(dw)} + \|H_{u,v,I,1}^{\sigma,\tau}\|_{L^s(dw)} \lesssim w(I)^{\frac{1}{s}-\frac{1}{2}}.$$

For $\varepsilon = 2$, if $\text{osc}_{r_s}(H_{\sigma,\tau}(f); I) > 0$, using Lemma 4.3 and (4.24), we obtain

$$\|E_{u,v,I,2}^{\sigma,\tau}\|_{L^s(dw)} \lesssim (\text{osc}_{r_s}(H_{\sigma,\tau}(f); I))^{-1} w(I)^{-1/2} \left(\int_I |H_{\sigma,\tau}(f) - a_{\sigma,\tau,I}(f)|^s dw \right)^{1/s} \lesssim w(I)^{\frac{1}{s}-\frac{1}{2}}.$$

If $\text{osc}_{r_s}(H_{\sigma,\tau}(f); I) = 0$, the same estimate is immediate from $E_{u,v,I,2}^{\sigma,\tau} = 0$. Meanwhile,

$$\|H_{u,v,I,2}^{\sigma,\tau}\|_{L^s(dw)} \lesssim w(I)^{\frac{1}{s}-\frac{1}{2}}.$$

Finally, for $\varepsilon = 3$, since

$$E_{u,v,I,3}^{\sigma,\tau} = e_{u,v,I}^{\sigma,\tau}, \quad H_{u,v,I,3}^{\sigma,\tau} = h_{u,v,I}^{\sigma,\tau},$$

we apply Lemma 4.3 and $w(I^*) \lesssim w(I)$ to see that

$$\|E_{u,v,I,3}^{\sigma,\tau}\|_{L^s(dw)} + \|H_{u,v,I,3}^{\sigma,\tau}\|_{L^s(dw)} \lesssim w(I)^{\frac{1}{s}-\frac{1}{2}}.$$

Consequently,

$$\|E_{u,v,I,\varepsilon}^{\sigma,\tau}\|_{L^s(dw)} + \|H_{u,v,I,\varepsilon}^{\sigma,\tau}\|_{L^s(dw)} \lesssim w(I)^{\frac{1}{s}-\frac{1}{2}}.$$

Step 3. Verification of (4).

By Lemma 4.6 with $A = C_1$,

$$\|\{\theta_{\sigma,\tau,I} \text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I)))\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \leq \|\{\text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I)))\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}.$$

Since $A_{\sigma,\tau,I,0}(f) = A_{\sigma,\tau,I,1}(f) = \theta_{\sigma,\tau,I} \text{osc}_{r_s}(g_\tau; B_\Gamma(z_I, C_1 \ell(I)))$, we have

$$\|\{A_{\sigma,\tau,I,\varepsilon}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}, \quad \varepsilon = 0, 1.$$

By Lemma 4.9,

$$\|\{A_{\sigma,\tau,I,2}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} = \|\{\theta_{\sigma,\tau,I} \text{osc}_{r_s}(H_{\sigma,\tau}(f); I)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}.$$

Finally, by Lemma 4.8,

$$\|\{A_{\sigma,\tau,I,3}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} = \|\{\theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}.$$

Thus, for every $\varepsilon \in \{0, 1, 2, 3\}$,

$$\|\{A_{\sigma,\tau,I,\varepsilon}(f)\}_{I \in \mathbb{D}_\Gamma}\|_{\ell^{p,q}} \lesssim \|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)}.$$

This completes the proof. $\square$

Theorem 4.11. *Assume that $N \geq 1$, $1 < p < \infty$ and $1 \leq q \leq \infty$. Fix $j \in \{1, \dots, N\}$. If $f \in OSC^{p,q}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in S^{p,q}(L^2(\mathbb{R}^N, dw))$. Moreover,*

$$(4.38) \quad \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)}.$$

Proof. Fix $s > 2$ and a closed Weyl chamber Γ . For fixed $\varepsilon \in \{0, 1, 2, 3\}$, $\sigma, \tau \in G$, and $u, v \in \mathbb{Z}^N$, by Lemma 4.10 (2)–(3) and the G -invariance of dw ,

$$\left\{ E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot) \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)} \right\}_{I \in \mathbb{D}_\Gamma}$$

are NWO families on Γ . Their supports are contained in I and I^* , respectively, and I^* is contained in a fixed enlargement of I .

For $\gamma \in G$, define

$$U_\gamma : L^2(\Gamma, dw) \longrightarrow L^2(\gamma\Gamma, dw), \quad U_\gamma g(x) := g(\gamma^{-1}x).$$

By the G -invariance of dw , U_γ is unitary. By Lemma 4.10 (2), the operator

$$\sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}} \right\rangle_{L^2(\mathbb{R}^N, dw)} E_{u,v,I,\varepsilon}^{\sigma,\tau}$$

vanishes on $L^2(\tau\Gamma, dw)^\perp$, and its range is contained in $L^2(\sigma\Gamma, dw)$. Hence, by Lemma 2.10 and Lemma 4.10 (4),

$$\begin{aligned} & \left\| \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}} \right\rangle_{L^2(\mathbb{R}^N, dw)} E_{u,v,I,\varepsilon}^{\sigma,\tau} \right\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))} \\ &= \left\| \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)} \right\rangle_{L^2(\Gamma, dw)} E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot) \right\|_{S^{p,q}(L^2(\Gamma, dw))} \\ &\lesssim \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)}. \end{aligned}$$

Choose $M_0 > 2N$. By Lemma 4.10 (1),

$$\sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| < \infty.$$

Since G is finite and $p > 1$, using an equivalent norm on $S^{p,q}$ when necessary, the operator series

$$T_f := \sum_{\varepsilon=0}^3 \sum_{\sigma,\tau \in G} \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}} \right\rangle_{L^2(\mathbb{R}^N, dw)} E_{u,v,I,\varepsilon}^{\sigma,\tau}$$

converges in $S^{p,q}(L^2(\mathbb{R}^N, dw))$, and

$$(4.39) \quad \|T_f\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))} \lesssim \sum_{\varepsilon=0}^3 \sum_{\sigma,\tau \in G} \sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| \|f\|_{OSC^{p,q}(\mathbb{R}^N)} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)}.$$

It remains to identify T_f with the commutator. Fix $r > p$. Since $\ell^{p,q} \subset \ell^r$, by Lemma 4.10 (4),

$$\left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^r} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)}.$$

For $m \in \mathbb{N}$, define

$$T_f^{(m)} := \sum_{\varepsilon=0}^3 \sum_{\sigma, \tau \in G} \sum_{u, v \in \mathbb{Z}^N} c_{u, v} \sum_{I \in \bigcup_{|k| \leq m} \mathbb{D}_{\Gamma, k}} A_{\sigma, \tau, I, \varepsilon}(f) \left\langle \cdot, \overline{H_{u, v, I, \varepsilon}^{\sigma, \tau}} \right\rangle_{L^2(\mathbb{R}^N, dw)} E_{u, v, I, \varepsilon}^{\sigma, \tau}.$$

By Lemma 2.10, for fixed $\varepsilon, \sigma, \tau, u, v$,

$$\left\| \sum_{I \in \bigcup_{|k| > m} \mathbb{D}_{\Gamma, k}} A_{\sigma, \tau, I, \varepsilon}(f) \left\langle \cdot, \overline{H_{u, v, I, \varepsilon}^{\sigma, \tau}} \right\rangle E_{u, v, I, \varepsilon}^{\sigma, \tau} \right\|_{S^r} \lesssim \left\| \left\{ A_{\sigma, \tau, I, \varepsilon}(f) \right\}_{I \in \bigcup_{|k| > m} \mathbb{D}_{\Gamma, k}} \right\|_{\ell^r} \longrightarrow 0.$$

Together with $\sum_{u, v} |c_{u, v}| < \infty$ and the preceding uniform ℓ^r -bound, this implies

$$(4.40) \quad \|T_f - T_f^{(m)}\|_{S^r} \longrightarrow 0.$$

By the independence of the defining exponent of $OS C^{p, q}$, as in the proof of Lemma 4.10, $f \in L_{\text{loc}}^s(\mathbb{R}^N, dw)$. Hence, for $\varphi, \psi \in C_c^\infty(\mathbb{R}^N)$, $f\varphi, \bar{f}\psi \in L^2(\mathbb{R}^N, dw)$, and the commutator form is well defined by

$$\langle [R_{\text{Dunkl}, j}, M_f] \varphi, \psi \rangle := \langle R_{\text{Dunkl}, j}(f\varphi), \psi \rangle - \langle R_{\text{Dunkl}, j}\varphi, \bar{f}\psi \rangle.$$

Let $R_{\text{Dunkl}, j}^{(m)}$ denote the smooth annular truncation obtained from the partition (4.22) by restricting the generations to $|k| \leq m$. By the construction in the proof of Lemma 4.3, together with (4.37) and the absolute summability in Lemma 4.10 (1),

$$(4.41) \quad \langle T_f^{(m)} \varphi, \psi \rangle = \langle R_{\text{Dunkl}, j}^{(m)}(f\varphi), \psi \rangle - \langle R_{\text{Dunkl}, j}^{(m)}\varphi, \bar{f}\psi \rangle.$$

By the $L^2(dw)$ truncated-kernel representation of the Dunkl Riesz transform,

$$R_{\text{Dunkl}, j}^{(m)} g \longrightarrow R_{\text{Dunkl}, j} g \quad \text{in } L^2(\mathbb{R}^N, dw), \quad g \in L^2(\mathbb{R}^N, dw).$$

Therefore, by (4.41),

$$\lim_{m \rightarrow \infty} \langle T_f^{(m)} \varphi, \psi \rangle = \langle [R_{\text{Dunkl}, j}, M_f] \varphi, \psi \rangle.$$

On the other hand, by (4.40),

$$\lim_{m \rightarrow \infty} \langle T_f^{(m)} \varphi, \psi \rangle = \langle T_f \varphi, \psi \rangle.$$

Thus, T_f is the bounded extension of $[R_{\text{Dunkl}, j}, M_f]$. Combining this identification with (4.39), we obtain

$$\| [R_{\text{Dunkl}, j}, M_f] \|_{S^{p, q}(L^2(\mathbb{R}^N, dw))} \lesssim \|f\|_{OS C^{p, q}(\mathbb{R}^N)}.$$

This completes the proof of Theorem 4.11. $\square$

5. PROOF OF LOWER BOUND

Definition 5.1. Let $1 \leq r < \infty$. For a measurable set $E \subset \mathbb{R}^N$ with $0 < w(E) < \infty$, set

$$(f)_{E, dw} := \frac{1}{w(E)} \int_E f dw$$

and

$$\text{osc}_{r, dw}(f; E) := \left(\frac{1}{w(E)} \int_E |f - (f)_{E, dw}|^r dw \right)^{1/r}.$$

For $0 < p < \infty$ and $0 < q \leq \infty$, define

$$\|f\|_{\text{OSC}_{dw}^{p,q}} := \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{1,dw}(f; Q)\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}}.$$

Lemma 5.2. Let $0 < p < \infty$, $0 < q \leq \infty$, and $1 \leq r < \infty$. For $f \in L_{\text{loc}}^r(\mathbb{R}^N, dx)$, set

$$\text{osc}_{r,dw}(f; Q) := \left(\frac{1}{w(Q)} \int_Q |f - (f)_{Q,dw}|^r dw \right)^{1/r}$$

and

$$\|f\|_{\text{OSC}_{dw}^{p,q}} := \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r,dw}(f; Q)\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}}.$$

Then

$$\|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)} \sim \|f\|_{\text{OSC}_{dw}^{p,q}},$$

where the implicit constants are independent of f .

Proof. By Lemma 2.1,

$$dw = \varpi dx, \quad \varpi \in A_\infty(\mathbb{R}^N, dx).$$

Moreover, both dx and dw are doubling measures. Hence dx and dw satisfy the A_∞ -relation in the sense of [12, Proposition 1.2]. Therefore,

$$\|f\|_{\text{Osc}^{p,q}(dx)} \sim \|f\|_{\text{Osc}^{p,q}(dw)}$$

by [12, Proposition 1.2].

By [12, (3.1) and Lemma 3.4], together with Lemma 2.8, these two oscillatory norms are equivalent, respectively, to the dyadic norms

$$\|f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)} \quad \text{and} \quad \|f\|_{\text{OSC}_{dw}^{p,q}}.$$

The assertion follows. $\square$

Lemma 5.3. Fix $j \in \{1, \dots, N\}$. For every $Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu$, let

$$r_Q := \sqrt{N} \ell(Q), \quad B_Q := B(c_Q, r_Q),$$

and define

$$\widehat{B}_Q := B(c_Q + 5r_Q e_j, r_Q).$$

Then

$$(5.42) \quad w(Q) \sim w(B_Q) \sim w(\widehat{B}_Q),$$

and there is a constant C_N , independent of Q , such that

$$Q \cup \widehat{B}_Q \subset C_N Q.$$

Moreover, for every $(x, y) \in Q \times \widehat{B}_Q$, the kernel $K_{\text{Dunkl},j}(x, y)$ does not change sign and

$$(5.43) \quad |K_{\text{Dunkl},j}(x, y)| \gtrsim \frac{1}{w(Q)}.$$

Proof. Since Q is a Euclidean cube,

$$B\left(c_Q, \frac{\ell(Q)}{2}\right) \subset Q \subset B_Q.$$

Hence the doubling property (2.2) gives

$$w(B_Q) \lesssim w\left(B\left(c_Q, \frac{\ell(Q)}{2}\right)\right) \leq w(Q) \leq w(B_Q),$$

and therefore

$$w(Q) \sim w(B_Q).$$

Since the distance between the centers of B_Q and $\widehat{B}_Q$ is $5r_Q$,

$$\widehat{B}_Q \subset 6B_Q.$$

Similarly,

$$B_Q \subset B(c_Q + 5r_Q e_j, 6r_Q) = 6\widehat{B}_Q.$$

The doubling property therefore yields

$$w(B_Q) \sim w(\widehat{B}_Q),$$

which proves (5.42).

For $y \in \widehat{B}_Q$,

$$\|y - c_Q\| \leq 6r_Q = 6\sqrt{N}\ell(Q).$$

Thus $Q \cup \widehat{B}_Q$ is contained in a fixed concentric enlargement $C_N Q$.

Finally, $Q \subset B_Q$, so Lemma 2.4, applied to B_Q , shows that $K_{\text{Dunkl},j}$ has a fixed sign on $Q \times \widehat{B}_Q$ and

$$|K_{\text{Dunkl},j}(x, y)| \gtrsim \frac{1}{w(B_Q)} \sim \frac{1}{w(Q)}.$$

□

Lemma 5.4. *Let $f \in L^1_{\text{loc}}(\mathbb{R}^N, dw)$ be real-valued, and assume that $[R_{\text{Dunkl},j}, M_f]$ is a compact operator on $L^2(\mathbb{R}^N, dw)$. For every $v \in \{1, \dots, 2^N\}$, there exist NWO sequences*

$$\{e_{Q,m}\}_{Q \in \mathbb{D}^v}, \quad \{h_{Q,m}\}_{Q \in \mathbb{D}^v}, \quad m = 1, 2,$$

such that, for every $Q \in \mathbb{D}^v$,

$$\text{osc}_{1,dw}(f; Q) \lesssim \sum_{m=1}^2 \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|.$$

Proof. Fix $v \in \{1, \dots, 2^N\}$ and $Q \in \mathbb{D}^v$. Set

$$r_Q := \sqrt{N}\ell(Q), \quad B_Q := B(c_Q, r_Q).$$

Since

$$B(c_Q, \ell(Q)/2) \subset Q \subset B_Q,$$

by (2.2),

$$w(B_Q) \simeq w(Q).$$

By Lemma 2.4, for

$$\widehat{B}_Q := B(c_Q + 5r_Q e_j, r_Q),$$

the kernel $K_{\text{Dunkl},j}$ has a fixed sign on $B_Q \times \widehat{B}_Q$ and

$$|K_{\text{Dunkl},j}(x, y)| \gtrsim \frac{1}{w(Q)}, \quad (x, y) \in Q \times \widehat{B}_Q.$$

Moreover,

$$\widehat{B}_Q \subset 6B_Q, \quad B_Q \subset 6\widehat{B}_Q,$$

so (2.2) implies

$$w(\widehat{B}_Q) \simeq w(Q).$$

Since $\widehat{B}_Q \subset B(c_Q, 6r_Q)$, there is a fixed $C_N \geq 1$ such that

$$\widehat{B}_Q \subset C_N Q, \quad w(C_N Q) \simeq w(Q).$$

Let α_Q be a dw -median of f on $\widehat{B}_Q$, and define

$$\begin{aligned} E_{Q,1} &:= \{x \in Q : f(x) < \alpha_Q\}, & F_{Q,1} &:= \{y \in \widehat{B}_Q : f(y) \geq \alpha_Q\}, \\ E_{Q,2} &:= \{x \in Q : f(x) \geq \alpha_Q\}, & F_{Q,2} &:= \{y \in \widehat{B}_Q : f(y) \leq \alpha_Q\}. \end{aligned}$$

Then

$$w(F_{Q,m}) \geq \frac{1}{2}w(\widehat{B}_Q) \simeq w(Q), \quad m = 1, 2.$$

For each $m = 1, 2$, $f(y) - f(x)$ has a fixed sign on $E_{Q,m} \times F_{Q,m}$, and

$$|f(x) - \alpha_Q| \leq |f(y) - f(x)|, \quad (x, y) \in E_{Q,m} \times F_{Q,m}.$$

Since $K_{\text{Dunkl},j}$ also has a fixed sign on $Q \times \widehat{B}_Q$, we obtain

$$\begin{aligned} \frac{1}{w(Q)} \int_{E_{Q,m}} |f - \alpha_Q| dw &\leq \frac{1}{w(Q)w(F_{Q,m})} \int_{E_{Q,m}} \int_{F_{Q,m}} |f(y) - f(x)| dw(y) dw(x) \\ &\lesssim \frac{1}{w(Q)} \left| \int_{E_{Q,m}} \int_{F_{Q,m}} (f(y) - f(x)) K_{\text{Dunkl},j}(x, y) dw(y) dw(x) \right|. \end{aligned}$$

Define

$$e_{Q,m} := w(C_N Q)^{-1/2} \chi_{F_{Q,m}}, \quad h_{Q,m} := w(Q)^{-1/2} \chi_{E_{Q,m}}.$$

By the off-diagonal kernel identity

$$K_{[R_{\text{Dunkl},j}, M_f]}(x, y) = (f(y) - f(x)) K_{\text{Dunkl},j}(x, y),$$

and $w(C_N Q) \simeq w(Q)$,

$$\frac{1}{w(Q)} \int_{E_{Q,m}} |f - \alpha_Q| dw \lesssim \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|.$$

Furthermore,

$$\text{supp } h_{Q,m} \subset Q, \quad \text{supp } e_{Q,m} \subset C_N Q.$$

For any fixed $s > 2$,

$$\begin{aligned} \|h_{Q,m}\|_{L^s(\mathbb{R}^N, dw)} &\leq w(Q)^{1/s-1/2}, \\ \|e_{Q,m}\|_{L^s(\mathbb{R}^N, dw)} &\leq w(C_N Q)^{1/s-1/2} \leq w(Q)^{1/s-1/2}. \end{aligned}$$

Hence, $\{e_{Q,m}\}_{Q \in \mathbb{D}^\nu}$ and $\{h_{Q,m}\}_{Q \in \mathbb{D}^\nu}$ are NWO sequences for $m = 1, 2$.

Finally,

$$|(f)_{Q,dw} - \alpha_Q| \leq \frac{1}{w(Q)} \int_Q |f - \alpha_Q| dw,$$

and therefore

$$\begin{aligned} \text{osc}_{1,dw}(f; Q) &\leq \frac{2}{w(Q)} \int_Q |f - \alpha_Q| dw \\ &= 2 \sum_{m=1}^2 \frac{1}{w(Q)} \int_{E_{Q,m}} |f - \alpha_Q| dw \\ &\lesssim \sum_{m=1}^2 \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|. \end{aligned}$$

This completes the proof of Lemma 5.4. $\square$

Theorem 5.5. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. Fix $j \in \{1, \dots, N\}$. Let

$$f \in L^1_{\text{loc}}(\mathbb{R}^N, dx).$$

If

$$[R_{\text{Dunkl},j}, M_f] \in S^{p,q}(L^2(\mathbb{R}^N, dw)),$$

then $f \in OSC^{p,q}(\mathbb{R}^N)$ and

$$(5.44) \quad \|f\|_{OSC^{p,q}(\mathbb{R}^N)} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

Proof. Assume first that f is real valued. Fix $\nu \in \{1, \dots, 2^N\}$. By Lemma 5.4, for $m = 1, 2$ there are NWO sequences

$$\{e_{Q,m}\}_{Q \in \mathbb{D}^\nu}, \quad \{h_{Q,m}\}_{Q \in \mathbb{D}^\nu},$$

such that

$$\text{osc}_{1,dw}(f; Q) \lesssim \sum_{m=1}^2 \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|.$$

By the quasi triangle inequality in $\ell^{p,q}$ and Lemma 2.9,

$$\begin{aligned} \left\| \left\{ \text{osc}_{1,dw}(f; Q) \right\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} &\lesssim \sum_{m=1}^2 \left\| \left\{ \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} \\ &\lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}. \end{aligned}$$

Summing over ν and using Definition 5.1, we obtain

$$\|f\|_{OSC^{p,q}_{dw}} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

By Lemma 5.2 with $r = 1$,

$$(5.45) \quad \|f\|_{OSC^{p,q}(\mathbb{R}^N)} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

For general f , let C denote complex conjugation on $L^2(\mathbb{R}^N, dw)$. Since $K_{\text{Dunkl},j}$ is real valued by its heat kernel representation,

$$CR_{\text{Dunkl},j}C = R_{\text{Dunkl},j}, \quad C[R_{\text{Dunkl},j}, M_f]C = [R_{\text{Dunkl},j}, M_{\bar{f}}].$$

Hence,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_{\text{Re } f}] &= \frac{1}{2}([R_{\text{Dunkl},j}, M_f] + C[R_{\text{Dunkl},j}, M_f]C), \\ [R_{\text{Dunkl},j}, M_{\text{Im } f}] &= \frac{1}{2i}([R_{\text{Dunkl},j}, M_f] - C[R_{\text{Dunkl},j}, M_f]C). \end{aligned}$$

Since conjugation preserves singular values,

$$\|[R_{\text{Dunkl},j}, M_{\text{Re } f}]\|_{S^{p,q}} + \|[R_{\text{Dunkl},j}, M_{\text{Im } f}]\|_{S^{p,q}} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}}.$$

By (5.45) applied to $\text{Re } f$ and $\text{Im } f$,

$$\|\text{Re } f\|_{OS^{p,q}(\mathbb{R}^N)} + \|\text{Im } f\|_{OS^{p,q}(\mathbb{R}^N)} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

Since

$$\text{osc}_1(f; Q) \leq \text{osc}_1(\text{Re } f; Q) + \text{osc}_1(\text{Im } f; Q),$$

the quasi triangle inequality in $\ell^{p,q}$ proves (5.44). This completes the proof of Theorem 5.5. $\square$

6. APPENDIX

Lemma 6.1. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. Then there exists a constant $C > 0$ such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in \mathcal{E}} Q$ we may choose a cube $Q_x \in \mathcal{E}$ of minimal volume such that $x \in Q_x$. Every $Q \in \mathcal{E}$ containing x is then a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix $t > 0$ and let $\mathcal{E}$ be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \# \mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\# \mathcal{E})^{1/s'},$$

so

$$\# \mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since $\mathcal{E}$ is arbitrary,

$$\#\left\{Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t\right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak ℓ^s estimate follows. This completes the proof of Lemma 6.1. $\square$

Lemma 6.2. *Assume that $N > 1$. Then the following statements hold.*

(1) *For every $f \in L^1_{\text{loc}}(\mathbb{R}^N)$, we have*

$$(6.46) \quad \|\nabla f\|_{L^N(\mathbb{R}^N)} \simeq \|f\|_{\text{osc}^{N,\infty}(\mathbb{R}^N)}.$$

(2) *If $1 \leq q < \infty$ and $f \in \text{OSC}^{N,q}(\mathbb{R}^N)$, then f is constant almost everywhere.*

(3) *If $1 \leq p < N$, $1 \leq q \leq \infty$, and $f \in \text{OSC}^{p,q}(\mathbb{R}^N)$, then f is constant almost everywhere.*

Proof. This is a well-known result due to Rochberg–Semmes. $\square$

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